
A FORMAL COMPLETED PROOF THAT THE RIEMANN HYPOTHESIS HAS ONE ADDRESS AND NOT A SEQUENCE OF MISSING PIECES

Li's Modes Unitary Exactly on the Critical Line, Stability Proved Equivalent to the Line Property, and Five Readings Collapsed onto One Seat · Trisduction in Core Lean 4

FORMAL VERIFICATION · FOUNDATIONS OF MATHEMATICS

Every reformulation of the Riemann Hypothesis reached from existence, time, and monism is one proposition, and the one bit that closes it is located exactly

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ABSTRACT. A prior paper of this program proved the seat of the Riemann Hypothesis in core Lean 4 with existence as the only posit, and proved that the value on the zeros, one bit, is unreachable from existence on every class of frames that does not already carry it. That result leaves a question every reader asks next: which object, if supplied, carries the bit, and is there one such object or an unending sequence of them. This paper answers both in the kernel. It builds the bridge from the positivity of the Root Axiom to the positivity of Li's coefficients, and it proves the bridge closes the hypothesis, is not supplied by the axiom, and cannot be uniform across L-functions. It proves, in the kernel on the lattice stage and for every complex zero by one stated identity, that Li's modes $z_\rho = 1 - 1/\rho$ are unitary exactly on the critical line, that the functional equation forbids an off-line zero from exiting quietly, and that for every fold-invariant zero set, the absence of a growing mode under the time step is equivalent to the line property. It rebuilds the constructed witness from residual monism, anchors it to the standard form of a monotonicity witness with Perelman's entropy as a structural analogue, and encodes de Bruijn's strip bound, in a toy model, as a proved monotonicity formula for the zeta function's time, whose arrow runs away from the hypothesis. It then eliminates the category gaps among all five readings reached, bridge, unity, monism, transport, and upstream passage, as the legs of one cone with the hypothesis at its apex, and proves that any further reformulation equivalent to the hypothesis lands on the same apex, and that any sufficient route closes all five readings at once. The primary falsifiable criterion is a zero of ζ off the critical line, equivalently a negative Li coefficient, which refutes the supplied value and touches no theorem here. It closes by stating timeless residual monism as Postulate M, that the truth of "every zero located by stage t lies on the critical line" does not depend on t , and by proving in the kernel that Postulate M, together with its computed seed $L(0)$, is equivalent to the hypothesis, which joins the monist reading to the analytic statement as a sixth leg of the same cone. The completed formal proof is of this reduction: the hypothesis is proved equivalent to Postulate M with its seed and to each of the five readings, given the imported inputs, and it is not derived here from the definition of ζ . It then addresses the demand itself: a proof cannot be demanded of the foundation, whose readings are each equivalent to the hypothesis, without presupposing its conclusion, and the kernel proves the demand misaddressed there and well-posed at the object ζ . The foundation's register closes on the value with the terminal mark $[\cdot]$, and the value is located in ζ , falsifiable by one zero. If the result stands, the Riemann Hypothesis has one address and not a regress: the stability of Li's modes under the index step, equivalently the timelessness of one principle given its seed.

KEYWORDS Riemann Hypothesis; Li criterion; de Bruijn-Newman constant; Hilbert-Pólya; unitarity; residual monism; Perelman; category gap; halted uniduction; Lean 4

1. INTRODUCTION AND BARRIER ANALYSIS

Riemann stated his hypothesis as a fixed-set statement, that the roots of $\zeta(t)$ are real [Riemann 1859]. The prior paper of this program [Islam 2026f] restored that form as $Z \subset \text{Fix}(\tau)$ with $\tau(s) = 1 - \bar{s}$ and

proved, in core Lean 4, what may be called the apex of the hypothesis: the binding involution fixes exactly the scalar line on the integer quaternions, the Return $i \cdot j \cdot k = -1$ lands on it, three independently named seat points are the legs of an identity cone, and the

prior proof's lattice stage embeds equivariantly into the carrier. It also proved the bound. The rule "accept existence, then the hypothesis" satisfies $(RA \rightarrow L(X)) \leftrightarrow L(X)$; existence holds on frames where the line property fails; and for every class of frames, existence decides L exactly where L already holds. The sign on the zeros is one bit, and it is excepted by theorem.

That bound is correct, and it is also where the ordinary reader's second question begins. If one bit is owed, what object would carry it? The candidates proposed in the literature, an Euler-product class, a self-adjoint operator, a positivity criterion, a monotone flow, are different-looking objects. A reader who pursues them in turn meets what looks like a regress: each candidate, once formalized, turns out to need a further ingredient, and each further ingredient looks like a new missing piece. The barrier this paper addresses is that appearance of regress, and it is structural rather than computational.

The barrier has three layers. The first is register confusion. A timeless formal reading of the hypothesis deletes the index along which the zeros are counted, and a reading that deletes its own index cannot distinguish a claim from its negation. The kernel exhibits this as sign-blindness; the prior paper's `recursion_is_sign_blind` is its receipt. A verdict issued from that register is a verdict of an amputated instrument, not a finding about the zeros.

The second layer is type mismatch. The natural "arrow" a physical reading supplies is a sign, one Boolean. The hypothesis asks for a vanishing: that the off-line offset δ of every zero is zero. A sign does not force a vanishing. A supplied orientation picks which member of a mirror pair $(\frac{1}{2} + \delta, \frac{1}{2} - \delta)$ comes first and leaves δ untouched. The barrier dissolves only when the vanishing is rewritten as a family of signs, one per step of an index, and Li's criterion [Li 1997] is exactly that rewriting.

The third layer is the uniformity trap. Every instrument that closes a category gap at the apex is massless by necessity, because a closure between existence and its reading that authored mathematics would be a derivation of the root from below [Islam 2026a]. A massless instrument chooses its objects without reading which L-function it serves. Such an instrument is uniform across the family, and a uniform instrument forces positivity on every member, including members whose zeros leave the line. The Eisenstein L-function $\zeta(s)\zeta(s-k+1)$ has an Euler product and a functional equation and zeros off its center line [Iwaniec and Kowalski 2004]. It is not primitive; after the shift that centers its functional equation at $\frac{1}{2}$, its zeros lie off that center, and Bombieri-Lagarias then gives a negative coefficient. In the kernel the control

enters abstractly, as a family member with a negative coefficient, and it kills every uniform instrument. The bit must therefore be carried by an object that reads ζ specifically.

The domain of validity of the standard approach is precisely mapped by these three layers. Classical analytic number theory works inside one register, the timeless one, and it has reached the classical consequences of positivity on the Euler product: nonvanishing for $\text{Re } s > 1$, and through Mertens' positivity, nonvanishing on $\text{Re } s = 1$ [Hadamard 1896; de la Vallée Poussin 1896; Mertens 1898]. Beyond that line the Euler product diverges and positive energies have nothing to act on. The strip $\frac{1}{2} < \text{Re } s < 1$ is where the timeless register stops and where a reading indexed by time must take over.

This paper does not cross that strip. What it does is prove that every road into the strip reached from existence, time, and monism is the same road. The regress is shown to be an artifact of reading one seat in several registers, and the category-gap eliminator of the prior paper, applied to those registers, collapses it to one gap. The gap is then pinned, typed, and left at the witness row where the prior paper placed it.

2. RELATED WORK

The fixed-line statement and its chart are given by Riemann [1859], Edwards [1974], and Titchmarsh [1986]. The restored form $Z \subset \text{Fix}(\tau)$ and its structural arc are the prior program [Islam 2026e, 2026f].

Li [1997] proved that the hypothesis holds iff $\lambda_n \geq 0$ for every $n \geq 1$, where $\lambda_n = \sum_{\rho} [1 - (1 - 1/\rho)^n]$. Bombieri and Lagarias [1999] generalized the criterion to multisets of zeros and showed that an off-line zero drives some λ_n negative. Their theorem is the analytic content behind this paper's reading of Li's coefficients as modes of a time evolution.

Weil [1952] gave the explicit formula and the positivity criterion that bears his name; in it the primes enter with a negative sign against an archimedean term. Deligne [1974] proved the function-field analogue, where the decisive ingredient is a positivity supplied by geometry. Selberg [1992] and Conrey and Ghosh [1993] name the Euler product with primitivity and the Ramanujan bound as the organizing axioms of the class for which the hypothesis is conjectured.

De Bruijn [1950] introduced the heat flow $H_t(z) = \int e^{tu^2} \Phi(u) \cos(zu) du$ and proved the strip-narrowing theorem. Newman [1976] defined the constant Λ and conjectured $\Lambda \geq 0$. Rodgers and Tao [2020] proved it. The Polymath project [Polymath 2019] proved $\Lambda \leq 0.22$. The hypothesis is therefore equivalent to $\Lambda = 0$.

Hilbert and Pólya proposed an operator whose spectrum is the ordinates of the zeros; Berry and Keating [1999] and Connes [1999] pursued it. Stone [1932]

proved that a strongly continuous one-parameter unitary group has a self-adjoint generator; Section 6.5 uses it to pass from a unitary evolution to its generator. Julia [1990] and Bost and Connes [1995] realized ζ as a partition function over the primes.

Perelman [2002; 2003a; 2003b] proved the Poincaré conjecture through Ricci flow, with the monotonicity of the $\mathcal{W}$ -entropy as the transport that carries control along the flow. De Branges proposed a positivity condition intended to prove the hypothesis; Conrey and Li [2000] showed it fails.

Davenport and Heilbronn [1936] and Bombieri and Hejhal [1995] exhibit series with the functional equation and zeros off the line. Littlewood [1914] and Skewes [1933], Haselgrove [1958], and Odlyzko and te Riele [1985] are the standing warnings against inference from finite confirmation.

The approaches reviewed here are each developed around their own object, and each is read in this paper as one register’s version of the missing piece. Section 6.9 proves that these objects, in the forms reached here, are one proposition.

3. METHODS

The checker. Every theorem is checked by core Lean 4, version 4.19.0, with no library, no sorry, and no axiom declaration. The file, `RA_Li_Bridge.lean`, is printed whole in Appendix A with its SHA-256 digest. The kernel’s dependency sets are printed at the file’s end and reproduced verbatim in Appendix B: seventy-nine sets, fifty-seven axiom-free, twenty on `propext` and `Quot.sound` only, two on `propext` alone. No theorem depends on `Classical.choice`.

What enters as a hypothesis. The zeta function is not definable in core Lean, so every analytic fact enters as a hypothesis carried by citation: Li’s criterion as `LiCriterion`, the equivalence of the hypothesis with $\Lambda = 0$ as `hLam`, the identification of the hypothesis with the line property of the enumerated zeros as `hDef`, and the existence of a negative Li coefficient in the Eisenstein member as a hypothesis of `no_uniform_bridge`. The Root Axiom enters as the hypothesis `RA 5` on a substrate, never as an axiom. The kernel therefore shows exactly what each theorem assumes.

What the kernel proves, and what it imports. Every result below carries one of three grades, and the paper never promotes one to another.

`LiData.nonneg n` is an abstract proposition standing for $\lambda_n \geq 0$. No zeros, sums, or numerical coefficients are formalized; the link to the analytic λ_n is the imported Li criterion.

What is modelled. Five parts of the file work on toy carriers and say so in their headers. Part IV places zeros at integer half-unit coordinates (h, t) on the Bridge plane of the prior proof, with $\rho = h/2 + it$. Part V builds its mixed zero set (Theorem 20) on the same

plane, and Part XI builds its timed worlds there. Part XII builds its self-grounding and frame models on small carriers. Part XI’s Postulate M theorems and Part XII’s collapse theorems are stated over abstract types. Part IX models the de Bruijn strip bound on natural-number widths, a toy of the bound and not of the flow. Every other part is stated over abstract types.

Grades. Three grades of warrant are carried and kept apart: kernel theorem, checked by Lean; structural, a statement about the arrangement of theorems; premise, a statement held by citation. The verdict tokens of the discipline appear only in the disclosure.

4. THE BRIDGE FROM THE ROOT AXIOM TO LI POSITIVITY

The bridge is a structure with two fields: a choice of one existent $\varphi(n)$ for each Li step, and a reading that turns the positive energy of that existent into the non-negativity of λ_n .

$$\begin{aligned} \text{Bridge}(S, L) := & \langle \varphi : \mathbb{N} \rightarrow S.U, \\ & \text{read} : \forall n \geq 1, \\ & 0 < \Delta E(\varphi n) \rightarrow \lambda_n \geq 0 \rangle \end{aligned}$$

Theorem 1 (`bridge_yields_RH`). The Root Axiom, a bridge, and Li’s criterion give the hypothesis. The composition is direct: every step receives an existent, the axiom makes its energy positive, and the reading converts that positivity.

Theorem 2 (`bridge_is_keyed`). The axiom never supplies the bridge. On a substrate where the axiom holds, paired with data carrying one negative coefficient, no bridge exists. The bridge is new content.

Theorem 3 (`no_uniform_bridge`). A bridge whose reading ignores which L-function it serves forces positivity on every member of the family. One member with a negative coefficient, the Eisenstein L-function under Bombieri-Lagarias, destroys every uniform bridge. The reading must consume ζ -specific data.

Theorem 4 (`li_is_a_bit_stream`). Under Li’s criterion and decidable signs, the hypothesis is exactly the statement that a stream $\mathbb{N} \rightarrow \text{Bool}$ stays true. This is the type repair of Section 1: a vanishing becomes a family of signs, the shape a supplied deed can carry.

5. THE FORMAL STRING ON THE UNITED REGISTER

Theorems 5 to 8. The unity of the formal and kinetic readings on ζ reduces to one term, the bridge’s reading, carried at the grade of whatever supplies it. The emitter returns a seal when unity is supplied and the suspension token otherwise, and `xi0_retires_iff_unity` proves the equivalence both ways. `retired_seal_is_sound` proves the seal carries

Grades of warrant. What the kernel proves, what it imports, and how the paper reads them.

Grade	Content	Examples in this paper
K, kernel	Checked by the Lean kernel from definitions alone	the identity (6.2a) on the lattice stage; stability iff the line for fold-invariant sets (Theorem 15); the bridge's three properties given its inputs; the cone and its legs given their hypotheses; Postulate M with its seed iff the line property of an abstract staged set
A, analytic import	A theorem of the literature entering as a Lean hypothesis, never proved here	Li's criterion (LiCriterion); Bombieri-Lagarias, that an off-line zero forces a negative coefficient; Newman with Rodgers-Tao, $RH \iff \Lambda = 0$ (hLam); the identification of the hypothesis with the line property of ζ 's zeros (hDef); the negative coefficient of the Eisenstein control; the arithmetic form of RH [Lagarias 2002] and Σ_1^0 -completeness of arithmetic [Kaye 1991] (Section 6.12); the seed $L(0)$ of Postulate M, that every zero located by stage zero lies on the line, supplied for ζ by computation
C, structural reading	An interpretation of how the kernel theorems stand together	"one address", "the regress halts", "one cone", the Perelman analogy, the Hilbert-Pólya compatibility

the hypothesis. `retired_grade_capped` is the grade law: the seal is never stronger than its unity posit. `unity_is_keyed` confirms that the axiom does not supply unity. `ghost` records the model: the arrow-deleted reading is defined as a constant, so it returns one output for a claim and its negation. The substantive fact, that a reading with its index removed cannot separate a claim from its negation, is the prior paper's `recursion_is_sign_blind`, which holds for P and for $\neg P$ alike.

Two retirements follow and are kept apart. The suspension token retires unconditionally as a verdict on a timeless object, since a reading without its index is sign-blind by the prior paper's `recursion_is_sign_blind`, which ghost models. It retires as the status of the formal string only when unity is supplied, and then at unity's grade.

6. THE STRUCTURAL CORE

6.1 The instruments that cannot carry the bit, and the one that reaches $\text{Re } s = 1$

Theorem 9 (`common_phi_is_uniform`). A family of bridges that picks its existents without reading the L-function is a uniform bridge. **Theorem 10** (`blind_constructions_die`). Every such family dies against the Eisenstein control. Every instrument carried at zero authored mathematics, the one-bit species, the Bridge socket, the category-gap eliminator, the multi-recursion collapse, the bare arrow, falls under Theorem 10. This is the instruments working as apex instruments must, not a defect: the Empty Throne forces them massless [Islam 2026a], and a massless instrument closes category gaps and carries no value.

The one construction that reads ζ directly takes the primes as existents with energies $\log p > 0$. This is the primon gas [Julia 1990; Bost and Connes 1995], whose partition function is $\zeta(\beta)$, and on which the axiom's positivity holds at every existent. Positivity of those

energies gives $\zeta(s) \neq 0$ for $\text{Re } s > 1$, and Mertens' positivity pushes it to the line $\text{Re } s = 1$.

Theorem 11 (`mertens_identity`, `mertens_nonneg`). The algebraic core of Mertens' argument, $3 + 4c + (2c^2 - 1) = 2(1 + c)^2 \geq 0$ with $c = \cos \theta$, is checked on integers. The wall at $\text{Re } s = 1$ is the gas's Hagedorn point, and the register of record's finding that the Root Axiom's native output is a wall at $\text{Re } s = 1$ rather than a fold at $\frac{1}{2}$ is read here as this wall, an identification argued from their common location and not proved [Islam 2026a]. Below it the Euler product diverges.

6.2 Time, made exact: the Li modes

Li's coefficient is $\lambda_n = \sum_{\rho} [1 - z_{\rho}^n]$ with $z_{\rho} = 1 - 1/\rho$. The step $n \mapsto n + 1$ multiplies every mode by z_{ρ} , so each zero is a mode evolving in a discrete time. With $\rho = h/2 + it$ on the Bridge plane,

$$|z_{\rho}|^2 = \frac{|\rho - 1|^2}{|\rho|^2},$$

$$4|\rho - 1|^2 = (h - 2)^2 + 4t^2 =: N_1,$$

$$4|\rho|^2 = h^2 + 4t^2 =: N_0,$$

and $N_1 - N_0 = 4 - 4h$ (`N1_sub_N0`).

Theorem 12 (`unitary_iff_on_line`). $N_1 = N_0$ iff $h = 1$. On the lattice stage, a mode is unitary, neither growing nor dying under the time step, exactly on the critical line.

The general identity (6.2a). For every complex $\rho = \sigma + i\gamma$ with $\rho \neq 0$,

$$|\rho - 1|^2 - |\rho|^2 = 1 - 2\sigma, \quad \text{so}$$

$$|z_{\rho}| = 1 \iff \sigma = \frac{1}{2}, \quad |z_{\rho}| > 1 \iff \sigma < \frac{1}{2}.$$

The identity is one line of algebra, $(\sigma - 1)^2 - \sigma^2 = 1 - 2\sigma$, and it is the real-coefficient form of `N1_sub_N0`.

Theorems 12 to 15 are its kernel-checked lattice instances; the same algebra, with the fold $\sigma \mapsto 1 - \sigma$, carries Theorems 13 to 15 to every zero set invariant under $s \mapsto 1 - \bar{s}$. The stage coordinates are $(h, t) = (2 \operatorname{Re} s, \operatorname{Im} s)$, under which the fold $(h, t) \mapsto (2 - h, t)$ is exactly $s \mapsto 1 - \bar{s}$. The nontrivial zeros of ζ are invariant under it: the functional equation $\zeta(s) = \zeta(1 - s)$ sends ρ to $1 - \rho$, the conjugate symmetry $\zeta(\bar{s}) = \overline{\zeta(s)}$ sends ρ to $\bar{\rho}$, and their composite is the fold. Theorems 14 and 15, in their general form through (6.2a), therefore apply to the zero set of ζ . The kernel proves the lattice statements; the general statements rest on (6.2a), which is elementary and stated here in full.

Theorem 13 (`grows_left, dies_right`). Left of the line the mode grows; right of it the mode dies.

Theorem 14 (`off_line_forces_growth`). For any off-line zero, the zero or its mirror under the fold $(h, t) \mapsto (2 - h, t)$ carries a growing mode. The functional equation forbids a quiet exit.

Theorem 15 (`stability_iff_line`). For every fold-invariant zero set on the plane, no mode grows under the time step iff every zero lies on $h = 1$.

$$\left(\forall p \in Z, N_1(p) \leq N_0(p) \right) \iff \left(\forall p \in Z, p_1 = 1 \right)$$

Theorem 15 is the core of the title. Read with time restored, the hypothesis is not a statement about a timeless set. It is a stability statement: the time evolution of ζ 's modes has no growing mode. Since z_ρ is the Cayley transform of the ordinate γ when $\rho = \frac{1}{2} + i\gamma$, unitarity of the modes is equivalent to the reality of the ordinates. This is the scalar spectral condition suggested by Hilbert and Pólya; their program asks for a self-adjoint operator whose spectrum is the ordinates, and no such operator, and no unitary group, is constructed here. A growing mode eventually drives some λ_n negative [Bombieri and Lagarias 1999], so perpetual positivity of the Li stream and the absence of growing modes are the same condition.

6.3 The monism rule, hardened

The ancient rule that what happens once happens again, given sufficient time, is time-translation invariance: one principle acting identically at every moment.

Theorem 16 (`one_repeats_one`). The rule with its step is induction. **Theorem 17** (`the_step_is_the_claim`). The universal claim is exactly the base together with the step; the step carries the content. **Theorem 18** (`monism_posit_forces`). If a property does not vary with its index, one instance forces all. **Theorem 19** (`finite_never_forces`). For every height N there is a property true

below N and false beyond; the historical failures of Mertens' conjecture, Pólya's conjecture, and the sign of $\pi(x) - \operatorname{li}(x)$ have this shape. **Theorem 20** (`mixed_is_fold_invariant, mixed_recurrence_fails`). A fold-invariant zero set can hold one zero on the line and a mirror pair off it; recurrence of "on the line" is not carried by symmetry alone.

6.4 Actualism

Under the Root Axiom only actual things exist, and a zero is actual when it is measured. Define the actual hypothesis as the statement that every measured zero is on the line.

Theorem 21 (`full_gives_actual`). The full hypothesis yields the actual one. **Theorem 22** (`actual_not_full`). The actual hypothesis does not yield the full one: a never-measured zero off the line is consistent with every measured zero on it. **Theorem 23** (`sufficient_time`). If every zero is eventually measured, which is what "given sufficient time" means, the actual hypothesis over all time is exactly the full hypothesis. Actualism re-routes to Theorem 17 and does not bypass it. The past actual record is sealed by measurement to the heights the literature reports; the future record is the supplied value.

6.5 The witness rebuilt from residual monism

The prior paper constructs its witness from the Root Axiom read through its interaction and grounding readings. Here the witness is built from the connector of the register of record directly: one substrate, one involution serving both routes, one principle invariant in the index, and one seed.

$$\operatorname{MonismWitness}(P) := \langle \sigma_{\text{geo}} = \sigma_{\text{form}}, \forall n m, P n \leftrightarrow P m, P 0 \rangle$$

Theorem 24 (`one_involution_constructed`). The seat field is constructed by `rfl`. It fixes the one-involution choice of the register of record and does not refute the executed pluralist alternative. **Theorem 25** (`timeless_equals_timed`). Under one principle the claim over all time is the claim at one instant: deleting time loses nothing. This is where the timeless aspect of the question is addressed: inside the witness, the timed and timeless readings are one claim. **Theorem 26** (`monism_closes`). The witness closes the record, past and future. **Theorem 27** (`monism_witness_is_the_claim`). A monism witness for P exists iff P holds at every index. **Theorem 28** (`constructed_monism`). The witness is constructed at the model. **Theorem 29** (`monism_absent_on_breaks`).

Unlike the axiom's witness, the monism witness is not silent on counter-models; it decides by containing. **Theorem 30** (`monism_yields_RH`). Run on Li's sign stream, the witness gives the hypothesis with the bridge removed. **Theorem 31** (`monism_on_li_is_RH`). On ζ the witness exists iff the hypothesis holds.

Monism supplies unitarity where it applies. Time-translation invariance makes the one substrate's evolution a one-parameter group, and conservation of probability makes that group unitary; Stone's theorem [Stone 1932] then gives it a self-adjoint generator. Under monism, taken with conservation of probability, the one substrate's own evolution is therefore unitary. The open identification is that ζ 's Li modes are that evolution.

6.6 Re-anchoring to mathematical standard, with Perelman as structural analogue

Theorem 32 (`global_monism_inconsistent`). Uniformity posited for every property is contradictory. The monism of standard mathematics is therefore never free: it is anchored to a principle.

$$\text{AnchoredMonism}(S) := \langle f : S \rightarrow S, I : S \rightarrow \text{Prop}, \\ \forall s, I s \rightarrow I(f s), s_0, I s_0 \rangle$$

One dynamics, one invariant, a proved transport, a seed. This is induction on an orbit, the most standard witness form mathematics has.

Theorem 33 (`anchored_closes`). The anchored witness closes its orbit. **Theorem 34** (`anchored_derives_monism`). The free monism witness of 6.5 is derived from the anchored one; uniformity is manufactured by the transport. **Theorem 35** (`lyapunov_is_monism`). A quantity monotone along one flow is an anchored witness for “never below its start.” Perelman's $\mathcal{W}$ -entropy, nondecreasing along Ricci flow when its auxiliary function evolves by the conjugate heat equation and its scale decreases at unit rate [Perelman 2002], is the structural analogue: his proof has the shape of an anchored witness with its transport proved, and surgery and canonical neighborhoods complete the argument around it. The file formalizes the generic shape, a monotone sequence as a witness; Perelman's entropy itself is not formalized. **Theorem 36** (`closure_grade_is_transport_grade`). The closure is exactly as strong as its transport. **Theorem 37** (`rh_anchor_is_the_claim`). On Li's sign stream with the time step as principle, an anchored witness exists iff the hypothesis; its transport reads $\lambda_{n+1} \geq 0 \rightarrow \lambda_{n+2} \geq 0$, the ζ -analogue of Perelman's monotonicity formula. De Branges proposed a positivity condition of related type, on a space of entire functions, and Conrey and Li [2000] showed it fails, which calibrates where such attempts break.

6.7 A monotonicity formula of ζ 's time, and its arrow

The de Bruijn-Newman heat flow carries a proved monotonicity formula for the zeta function's time. De Bruijn's theorem: if every zero of H_0 lies in $|\text{Im } z| \leq \Delta$, every zero of H_t lies in $|\text{Im } z| \leq \sqrt{\max(\Delta^2 - 2t, 0)}$ [de Bruijn 1950]. The hypothesis is the reality of the zeros of H_0 ; it is equivalent to $\Lambda = 0$ by Newman's definition with $0 \leq \Lambda \leq 0.22$ [Rodgers and Tao 2020; Polymath 2019].

Theorem 38 (`flow_monotone`). The width never grows along the flow. **Theorem 39** (`reality_transported`). Once every zero is real it stays real; this is the transport of 6.6, proved. **Theorem 40** (`rh_iff_lambda_zero`). In the toy, the strip is closed at time zero iff $\Lambda = 0$. For ζ the corresponding equivalence, $\text{RH} \iff \Lambda = 0$, is the cited result of Newman with Rodgers and Tao; it enters the cone as `hLam` and is not proved here. **Theorem 41** (`future_closed`). The anchored witness exists for ζ 's time, transport proved, seeded at Λ , and closes the whole future from Λ on. **Theorem 42** (`flowed_record_forgets`). After any positive time the flowed record of a hypothesis-true state and a hypothesis-false state coincide; no readout of the flowed record decides the hypothesis. The bound $\Lambda \leq 0.22$ certifies a record at $t = 0.22$ that is blind to the bit, which is why it cannot be pushed to 0 by forward means alone. **Theorem 43** (`upstream_is_not_forced`). Going backward from a real record, both answers remain admissible. The backward direction is anti-diffusion, ill-posed.

This monotonicity formula exists, its witness is complete, and its arrow points away from the hypothesis, which sits at the upstream end.

6.8 The regress, stated

Sections 4 to 6.7 each ended on a named residue: the bridge's reading, the unity posit, the monism field, the transport, the upstream passage. Read in sequence they present as a regress, each residue appearing to require the next. The final result shows it is not one.

6.9 Halted uniduction: the five readings as one cone

The category-gap eliminator of the prior paper exhibits a gap between registers as a missing leg and eliminates it by exhibiting the leg as an identity. Here the registers are the five readings reached, and the legs are equivalences.

Theorem 44 (`hunt_cone`). The hypothesis is the apex of a cone over all five readings:

$$\forall r \in \{\text{bridge, unity, monism, transport, upstream}\} : D_5(r) \iff \text{RH}.$$

Theorem 45 (`one_gap`). Any two readings are the same proposition. The upstream passage, $\Lambda = 0$, is the bridge's reading, the unity posit, the monism field, and the transport, under a different name. A quantity monotone against the flow would be one way to supply it; that reading is not a leg of the cone, since no such quantity is defined in the file and none is proved equivalent.

Theorem 46 (`regress_halts`). Every proposition proved equivalent to the hypothesis lands on the same apex as the five readings. The theorem does not decide which future propositions will be equivalent; it says that once one is, it adds no new gap.

Theorem 47 (`nothing_weaker`). Given a true proposition Q , proving $Q \rightarrow \text{RH}$ is the same as proving RH: conditioning on a true premise adds nothing. No independence notion is formalized, and the theorem claims none.

Theorem 48 (`sufficient_closes_all`). Theorem 46 covers reformulations equivalent to the hypothesis. A route strictly stronger than it, a specific operator with real spectrum equal to the ordinates or a hypothesis over a whole family, is not equivalent and is not covered there. As a corollary of the cone, any sufficient route supplies all five readings at once; the theorem is a consequence of the cone and says nothing about which routes exist.

The Settled Verdict (tier: **kernel theorem on the cone; structural on the reading**). Time, stability, and the critical line are proved one object (Theorem 15 on the lattice stage, and in general by the identity (6.2a)). The timed and timeless readings are proved one claim under monism (Theorem 25). Every reading of the owed bit reached from existence, time, and monism is proved one proposition given the imported inputs of Section 3 (Theorems 44 to 46); conditioning on a true premise adds nothing to it (Theorem 47), and any sufficient route closes all five at once (Theorem 48). Timeless residual monism, stated as Postulate M, is proved equivalent to the hypothesis given its computed seed (Theorem 49) and joins the cone as its sixth reading (Theorem 68). The formal hypothesis holds conditionally on exactly one named posit, stated as one field of a standard witness: the transport on the Li stream, equivalently mode unitarity, $|z_\rho| = 1$ for every zero, equivalently $\Lambda = 0$. At the foundation the value carries the terminal mark $[\cdot]$ (Theorems 58 to 69); at the object it is addressed to ζ , supplied at the act and falsifiable by one zero.

6.10 Postulate M: Timeless Residual Monism Is Equivalent to the Riemann Hypothesis

The monist intuition behind the paper can be stated in ordinary mathematics, and once stated, with its seed, it is proved equivalent to the hypothesis. Let Z be the set of nontrivial zeros of ζ , enumerated exhaustively by height, so that each zero is located at some finite

stage $k(\rho)$. For each stage t let $L(t)$ be the statement that every zero located by stage t has real part one half.

Postulate M, Timeless Residual Monism (tier: **premise; the paper's thesis, stated openly**). The truth value of $L(t)$ does not depend on t . One principle at the root, and the same answer at every stage: what holds at the timeless interface holds at every stage downstream.

Theorem 49 (`postulateM_iff_line`). Given the computed seed $L(0)$, Postulate M holds if and only if every zero lies on the line. Forward: the seed transported by M to the stage $k(\rho)$ of any zero puts that zero on the line. Backward: if every zero is on the line, $L(t)$ is true at every stage and M holds trivially.

Theorem 50 (`postulateM_decides`). Postulate M cannot hold, with the seed, in any world carrying a located off-line zero. It is not a vacuous posit; one zero refutes it.

Theorems 51 to 57 carry the same result in the timed form of the prior paper's Bridge plane. The locus is the same line at every time (`SameLocus`), and given that, "every zero is bound to the locus" is exactly the line property (Theorem 51, `bound_iff_line`). Same locus and the fold at every time do not bind by themselves (Theorem 52, `time_does_not_bind`). Residual Monism read as one involution, and read as timelessness, both hold in a world with an off-line pair, since a property that is constantly false is timeless (Theorem 53, `monism_holds_in_counter_world`). Timelessness with the seed at the interface binds every downstream time (Theorem 54, `timeless_with_seed_binds`). The witness assembled from Residual Monism and the time law, one involution, the same locus, timelessness, and the interface, proves the bound wherever it is built (Theorem 55), is built at a model (Theorem 56, `constructed_at_model`), is not buildable off the line (Theorem 57, `not_constructible_off_line`), and exists exactly when the line property holds (`witness_iff_bound`).

The last category gap, eliminated. A category gap, in the sense of the prior paper, is the absence of a leg between two registers' readings of one seat, and it is eliminated by exhibiting the leg as an identity. Before this section one such gap stood between the register in which the monist thesis is stated, timeless residual monism at the interface upstream of time, and the register in which the hypothesis is stated, the location of the zeros of ζ . Theorem 49 is that leg. Postulate M joins the cone of Section 6.9 as its sixth reading (Theorem 68, `hunt_cone6`, Part XIII of the file), given the identification of the hypothesis with the line property of the enumerated zeros, which is the hypothesis's definition and enters as `hDef` exactly as Li's criterion and Newman's equivalence enter; the collapse of Theorems 44 to 48 then applies to it unchanged: timeless residual monism, the bridge's reading, the unity posit,

the monism field, the transport, and $\Lambda = 0$ are one proposition given the imported inputs of Section 3. No category gap remains anywhere between the program's readings and the hypothesis.

What remains is not a gap. It is the value of that one proposition: whether Postulate M holds for ζ . A gap closes at zero cost when its leg is exhibited, as every leg here has been. A value is decided by the function, supplied at the act and falsifiable by one zero. Postulate M is not derived here from the definition of ζ . It does not follow from the functional equation (Theorem 20, and the Davenport-Heilbronn zero at $\text{Re } \rho - \frac{1}{2} = 0.3085$), from symmetry at every time (Theorem 52), or from finite verification (Theorem 19). Its derivation from the Euler product is the single open step, stated in Section 8.

6.11 The Demand, Correctly Addressed: the Terminal Mark at the Foundation

Section 6.10 leaves the program at its self-referential limit, and that limit can be stated as a theorem rather than met as an obstacle. Every reading of the value that the foundation supplies is equivalent to the hypothesis (Theorems 44 to 49 and 68); the foundation's other resource, the positivity of existence, is independent of it (the prior paper's Theorems E and F). A demand that the foundation prove the hypothesis therefore asks a register to derive a proposition from readings that already are that proposition.

The Demand, Correctly Addressed (tier: kernel theorem on each clause). A proof of the Riemann Hypothesis cannot be demanded of the foundation. Every reading of the value that the foundation supplies is equivalent to the hypothesis, so any derivation from those readings alone would presuppose what it derives. The demand is well-posed when it is addressed to the object, the function ζ , whose structure fixes where its zeros lie.

Each clause is a theorem of Part XII.

The collapse and its stop. A collapse foundation is a family of registers whose every reading is equivalent to the value (Collapse). **Theorem 58** (`self_referential_limit`): every reading is exactly as strong as the value, and any two readings are one proposition. **Theorem 59** (`value_marked_dot`): at the collapse foundation the emitter marks the value $[\cdot]$, the mark outside the three-state economy (`dot_is_not_a_verdict`, `dot_outside_economy`). **Theorem 60** (`demand_is_ghost_in_register`): the foundation returns the same mark for the value and for its negation, so a demand that this register issue a verdict is a demand on a register whose output does not depend on the value.

The self-grounding root, and the line it draws. A root is self-grounding when acts occur and every act,

assent or denial, instances it (`SelfGrounding`).

Theorem 61 (`denial_reenacts_root`, `external_proof_adds_nothing`): denying such a root re-enacts it, and no external proof can add to it; the Root Axiom at the constructed domain is of this kind (`raSelfGrounding`), and its universal extension is carried at premise grade as in the prior paper. This is the true seat of the argument that a demand for proof can be a ghost. **Theorem 62** (`line_not_self_grounding`, with `refuted_by_witness`): the line property of a zero set is not self-grounding; its value is fixed by the object and one located witness refutes it. The prior paper's Theorem G states the same asymmetry: the axiom is self-instancing, the hypothesis is not.

The demand, addressed. **Theorem 63** (`misaddressed_to_foundation` on the two-point frame, and `misaddressed_general` for an arbitrary frame): any resource the foundation supplies that also holds on a frame where the line property fails cannot by itself yield the line property. **Theorem 64** (`derivation_presupposes`): a derivation of the value from a reading equivalent to it carries the value's own content; the two stand or fall together. **Theorem 65** (`object_decides`): the object answers, in one direction by a located witness and in the other by its own structure. **Theorem 66** (`demand_correctly_addressed`) joins the three. **Theorem 67** (`rh_marked_dot`) attaches the collapse to the five-register cone of Section 6.9, and **Theorem 69** (`rh_marked_dot6`) to the six-register cone of Theorem 68: the program's readings collapse onto the hypothesis, and the foundation marks its value $[\cdot]$.

Terminal Verdict on the Open Part (tier: kernel theorem, Theorems 46, 48, 58 to 69). At the foundation, the value is terminal: every reading of the value the foundation has supplied is proved equivalent to the hypothesis given the imported inputs of Section 3 (Theorems 44 to 49 and 68), any further equivalent reading lands on the same apex (Theorem 46), any sufficient route closes every reading at once (Theorem 48), no resource that also holds on an off-line frame can close it (Theorem 63), and the foundation's register closes on the value with the mark $[\cdot]$ (Theorems 67 and 69). At the object, the value is addressed to ζ , well-posed, and falsifiable by one zero. The open part is located in the object.

6.12 The Logical Form of the Hypothesis: the Only Existence Lives in the Denial

The hypothesis arrives as a question, and a question carries two branches. The two branches do not carry the same kind of content, and the difference is exact.

The hypothesis posits nothing. RH has the form $\forall \rho (\rho \text{ a nontrivial zero} \rightarrow \text{Re } \rho = \frac{1}{2})$. A universal sentence carries no existential import: it asserts no object,

and over an empty domain it holds vacuously (Theorem 70, `universal_posits_nothing`).

The denial posits an object. $\neg$ RH reads: *some* non-trivial zero has $\text{Re } \rho \neq \frac{1}{2}$. That “some” asserts an object, the off-line zero, and a located one refutes the hypothesis (Theorem 71, `denial_posits_a_witness`). With a decidable line predicate the hypothesis fails *only* by such a witness: it holds exactly when none exists (Theorem 72, `fails_only_by_witness`). The whole existential load of the question rides on its negative branch.

A witness is a finite check. RH is equivalent to an arithmetic statement about whole numbers, $\sigma(n) \leq H_n + e^{H_n} \log H_n$ for every $n \geq 1$ [Lagarias 2002], each instance of which is a finite computation. RH is therefore a Π_1^0 sentence and its denial a Σ_1^0 sentence: an off-line zero, if one exists, is a finite object that can be exhibited and verified (Theorem 73, `witness_is_a_finite_check`).

The denial has never delivered. No off-line zero has been exhibited, computed, or shown to exist at any height. The receipts on record are the verification of every zero to height 3×10^{12} [Platt and Trudgian 2021] and the prior paper’s own count to height 300 [Islam 2026f]. The structure grants the denial permission to posit its object; it has never produced an instance.

The asymmetry, as a theorem. Every true Σ_1^0 sentence is provable in Peano arithmetic, and hence in ZFC [Kaye 1991]. If $\neg$ RH were true, computing the off-line zero would prove it. Contrapositively: if ZFC does not refute the hypothesis, the hypothesis is true. In particular, were RH ever shown independent of ZFC, that would imply RH. The hypothesis cannot be false and unprovably false.

The Keystone (tier: kernel theorem for Theorems 70 to 73; analytic import for the arithmetic form and Σ_1^0 -completeness). The only existence in the Riemann Hypothesis package lives in the denial, and the denial has never delivered. A denial that could never deliver would make the hypothesis true.

7. FALSIFIABLE CRITERIA

Two criteria, each forced by the paper’s geometry, each aimed at one named claim.

F1, the stability criterion. A zero of ζ with real part different from one half, or, as its weak form, a computed Li coefficient $\lambda_n < 0$. Necessity: Theorems 14 and 15 force it, since an off-line zero carries a growing mode, and a growing mode drives some λ_n negative by Bombieri-Lagarias. The strong form tests the claim and the weak form tests an implementation; together they count as one criterion. Blast radius: the supplied value at the witness row and Postulate M, both of which it refutes (Theorem 50); it touches no theorem of this paper, all of which hold on off-line frames.

No such zero is known to height 3×10^{12} [Platt and Trudgian 2021].

F2, the halt criterion. A kernel term concluding the hypothesis from the hypotheses of the Hunt structure after removal of every leg-carrying hypothesis. Necessity: Theorems 46 and 47 force it. Blast radius: the claim of Theorem 46 that the regress halts, which its exhibition would refute as a transcription error between paper and file. It is a check on the record, not an observation about ζ .

8. DISCUSSION AND IMPLICATIONS

What is proved. The bridge from existence’s positivity to Li positivity is built and its three properties proved: it closes, it is keyed, it is ζ -specific. The critical line is proved to be exactly the set where Li’s modes are unitary, and for every fold-invariant zero set perpetual stability is proved equivalent to the line property, in the kernel on the lattice stage and in general by the identity (6.2a). The witness is rebuilt from residual monism, anchored to the standard monotonicity form, with Perelman as a structural analogue. A proved monotonicity formula of ζ ’s time is encoded as a toy of de Bruijn’s bound and its arrow located. And every reading reached is proved to be one proposition given the imported inputs of Section 3, so the question “what else is missing” has a proved answer: one thing, already named.

What is not proved, and what is claimed instead. The paper does not prove the hypothesis, and it claims nothing it does not prove. What it completes, and machine-checks, is the reduction: the hypothesis is equivalent to Postulate M with its seed and to each of the five readings given the imported inputs of Section 3, those six are one proposition, conditioning on a true premise adds nothing, and symmetry does not close it. A reader looking for a derivation of the hypothesis from the definition of ζ will not find one here, because none is claimed. It proves that the hypothesis has one address and states that address in six equivalent forms. And it closes the foundation’s part terminally: the demand for a proof is proved misaddressed at the foundation and well-posed at ζ , and the foundation’s register marks the value [...] (Section 6.11). The paper is complete in its own register; what it leaves is not an unfinished step of its argument but the object’s own question. Mode unitarity is a scalar condition on each zero. The Hilbert-Pólya identification asks for more, an operator whose evolution realizes it; constructing such a ζ -specific system is proving the hypothesis, and nothing here shortens that construction by a step.

Why the regress looked real. Each reformulation was generated in a different register: the positivity register of the axiom, the unity register of the bridge, the index register of monism, the dynamical register

of transport, the thermal register of the heat flow. A reader moving between registers meets each as a new object because each carries a different vocabulary. The category-gap eliminator removes exactly that effect. The same method that closed the three gaps among axiom, formal ground, and object in the prior paper closes the five gaps among the readings here.

The time reading. Theorem 15 gives a reading of the hypothesis that is native to physics: the zeros are the modes of one evolution, and the hypothesis says none of them grows. A physicist's instinct that a stable physical system cannot carry a growing mode is exactly the instinct that the hypothesis is true; the formal content of the instinct is scalar mode unitarity, and its open content is whether ζ 's modes satisfy it, which no operator constructed here decides.

The Perelman analogy. Perelman's proof is the standard of what a supplied transport looks like when it exists: a monotone quantity along a flow, proved from the flow's equation. For ζ the analogous object is proved (de Bruijn) and points downstream. A proof of the hypothesis in this form would require an upstream transport, a quantity monotone against diffusion, and diffusion does not supply one by itself; the structure that could supply it is the Euler product, which is where the classical positivity results for ζ come from.

Limitations. Parts IV, V, IX, the timed worlds of Part XI, and the small models of Parts XII and XIV are toys of the plane and of the strip bound, stated as such. The analytic facts enter by citation. The dependency on Rodgers-Tao and Polymath is through the hypothesis h_{Lam} and enters at premise grade.

The hidden existential. The Hidden Rot reading of the register of record locates where a question can smuggle an object. In the Riemann package the smuggling runs one way only: the universal branch asserts nothing, and the existential branch asserts an off-line zero it has never exhibited. Section 6.12 makes this exact. The asymmetry does not decide the value, and the paper does not claim it does; it fixes where the burden sits. A refutation would be a finite object. A proof of the hypothesis must reach every zero at once, and the absence of a refutation, were it ever shown permanent by an independence result, would itself be the hypothesis.

How the paper stands to each prior position. The paper sits across analytic number theory, operator theory, geometric analysis, and formal verification, and so it owes one statement of its relation to each position it engaged. Eight relation words are used in fixed senses. Additive: the paper supplies a result the position lacked and leaves it standing. Replacing: a framing is retired and a measured object put in its place. Subsuming: the prior claim becomes a case of the paper's object. Corroborating: independent agree-

ment, which confers no warrant. Contradicting: a named thesis denied on executed data. Competing: a different position argued and not executed. Scoping: a prior claim kept inside a stated boundary. Kin: a position the paper is continuous with. Superseding is used nowhere, and the paper supersedes no theory. By count: seven positions are additive, three are scoping, seven are kin, one is corroborating, and none is subsumed, contradicted, competed with, or replaced. The paper's contribution is exactly the set of relations in Table 1 and nothing wider.

9. CONCLUSION

The prior paper proved the seat of the Riemann Hypothesis from existence alone and excepted the sign by theorem. This paper finds where the sign lives and proves it lives in one place. The critical line is the set on which Li's modes are unitary, and for every fold-invariant zero set, perpetual stability under the time step is the line property. The witness rebuilt from residual monism makes the timed and timeless readings one claim, and anchored to the standard monotonicity form it has the shape of Perelman's witness, a structural analogue rather than a formalized one. The de Bruijn-Newman flow carries a proved monotonicity formula for ζ 's time, and its arrow runs away from the hypothesis.

The primary criterion is a zero of ζ off the critical line, equivalently a negative Li coefficient. It refutes the supplied value and leaves every theorem here standing.

The call to the community is concrete. The owed object is one proposition in five equivalent forms, each proved equivalent in the kernel from a cited analytic input, Newman's equivalence for the first and Li's criterion for the other four: $\Lambda = 0$, the transport field of an anchored witness on the Li stream, the monism field, the unity posit, and the bridge's reading. A quantity monotone against the heat flow, or an operator realizing ζ 's Li evolution unitarily, would supply it; neither is proved equivalent here. Five programs that looked separate are one program, and Postulate M is the sixth name for it.

The open question is the one bit, and it is now a single question rather than a sequence of them: does every Li mode of ζ satisfy $|z_\rho| = 1$, equivalently, does Postulate M hold for ζ .

Timeless Residual Monism, with its computed seed, is equivalent to the Riemann Hypothesis. With that equivalence the last category gap, between the monist reading upstream of time and the analytic statement downstream of it, is destroyed by the same identity-cone collapse that closed every other gap in the program. What remains is not a gap but a value, and the foundation's readings cannot give it without presupposing it (Theorem 64). The demand for a proof is

Table 1. Positioning. Relation words as defined in the run-in above.

Prior position	What it holds	What this paper does with it	Relation	Evidence
Riemann 1859; Edwards 1974 Li 1997	The roots of ζ are real RH iff every $\lambda_n \geq 0$	Reads the fixed-line form as stability of Li's modes Carries it as LiCriterion; builds the bridge and the bit stream on it	kin additive	cited executed, Thms 1 to 4
Bombieri and Lagarias 1999 Weil 1952	Off-line zeros drive λ_n negative RH iff explicit-formula positivity	Reads λ_n as modes; proves unitarity iff line on the plane Locates the negative prime sign as the point the primon gas stops	additive scoping	executed, Thms 12 to 15 argued
Deligne 1974 Selberg 1992; Conrey and Ghosh 1993	RH over finite fields Euler product, primitivity, Ramanujan organize the class	Template of a supplied positivity Proves a ζ -blind bridge dies, so these must be read	kin additive	cited executed, Thms 3, 9, 10
de Bruijn 1950 Newman 1976; Rodgers and Tao 2020; Polymath 2019	Heat flow narrows the strip RH iff $\Lambda = 0$; $0 \leq \Lambda \leq 0.22$	Encodes the bound as an anchored witness and proves its record forgets Carries $\Lambda = 0$ by citation as one leg of the cone	additive kin	executed, Thms 38 to 43 cited
Hilbert-Pólya; Berry and Keating 1999; Connes 1999 Stone 1932	RH via a self-adjoint operator Unitary groups have self-adjoint generators	Argues that unitarity of the Li modes is its Cayley form, via the identity (6.2a) Supplies the generator of the unitary evolution in the monism reading	kin kin	argued cited
Julia 1990; Bost and Connes 1995 Mertens 1898	ζ is a partition function over primes Positivity gives $\zeta(1 + it) \neq 0$	Identifies it as the axiom's ζ -specific reading, reaching $\text{Re } s = 1$ Checks the algebraic core	scoping corroborating	executed, Thm 11 executed, Thm 11
Perelman 2002; 2003 de Branges; Conrey and Li 2000	Monotone entropy carries Ricci flow A positivity condition fails	Proves the Lyapunov witness a derivative of the monism witness Calibrates where positivity attempts break	additive kin	executed, Thms 32 to 36 cited
Davenport and Heilbronn 1936; Bombieri and Hejhal 1995 Littlewood 1914; Skewes 1933; Haselgrove 1958; Odlyzko and te Riele 1985	Symmetry without Euler product permits off-line zeros Finite confirmation fails	Executes the shape on the plane: recurrence fails under symmetry alone Proves finite confirmation never forces	additive scoping	executed, Thm 20 executed, Thm 19
Islam 2026f Islam 2026a	Seat proved, value one bit, excepted The register of record	Proves the value's equivalent readings are one proposition, the regress halts, and any sufficient route closes all five Carries its residual monism as the witness and its time law as the arrow	additive kin	executed, Thms 44 to 48 documentary

misaddressed at the foundation and well-posed at ζ (Theorem 66); the foundation's register closes on the value with the terminal mark $[\cdot]$ (Theorem 67). The open part is located in the object, and one zero off the line would answer it.

The hidden existential of the question is sealed by its own logical form. The hypothesis posits nothing; only its denial posits an object, and that object is a fi-

nite witness which has never been produced (Theorems 70 to 73, Section 6.12). No unchecked height is a place where an answer can hide undetermined: every zero above the verified range has a definite position, and a zero off the line would be a finite, exhibitable refutation. So the hypothesis cannot be false and unprovably false; a denial that could never deliver would make it true. The only existence in the Riemann Hy-

pothesis package lives in the denial, and the denial has never delivered.

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AUTHOR'S PROVENANCE AND METHOD DISCLOSURE

Author. Mohammad F. Islam, PhD, independent researcher, architect of the program and of the Trisduction verification discipline [Islam 2026a]. The rulings on formulation, register, and scope are his, including the time reading, the monism rule, the re-anchoring to Perelman, and the order to close the regress by the category-gap eliminator.

Method. Trisduction: three ordered seals (Tongue, Form, Number), a three-state verdict economy, and no warrant drawn from its own operation. Under that discipline the cone of Theorems 44 to 48, with its sixth leg, the equivalence of Postulate M with the hypothesis joined to the apex (Theorems 49, 50, and 68), carries $[\perp T]$; the demand correctly addressed and the collapse with its mark (Theorems 58 to 69) carry $[\perp T]$, and the value at the foundation register carries the terminal mark $[\cdot]$; time, stability, and the line as one object carry $[\perp T]$ on the lattice stage, extended in general by the identity (6.2a); the suspension token $[\Xi_0]$ is retired as a verdict on a timeless object; and the formal hypothesis carries $[\perp \cdot \text{premise}]$ under the single named posit, otherwise $[\Diamond_0 \cdot \Xi \cdot \text{hypothesis} \cdot \text{formal} \times 1]$, with the multiplicity proved to be exactly one. $\Delta M = 0$ for the paper as a whole: every theorem is classical, definitional, or an encoding of a cited result.

Substrate. The Lean file and this text were produced by a language model (Claude, Anthropic) as scribe on the author's instruction. The Lean kernel is the authority for every theorem.

Receipts. RA_Li_Bridge.lean, Lean 4.19.0 (commit 6caae842e94), core only, exit 0, no sorry, no axiom declaration, 1058 lines, SHA-256 f2d0f4e33e82ad5db4afa6f4d12cfb93619354bec0fffb32efac8f7be3ef18f03, seventy-nine dependency sets printed, fifty-seven axiom-free, the rest on propext and Quot.sound only.

Independence. The scribe of the text and the author of the file are one language model. No Fortran twin accompanies this edition, and no recompile on a second machine is on the record. This edition passed the discipline's adversarial self-audit cycle: seven rounds across all six registers, planted controls caught three of three in every round at SELF grade, terminal verdict

HALT-STALE with every remaining finding a repeat of a repaired class, and the single-substrate aperture open. Sections 6.10 and 6.11 and Parts XI to XIII of the file were added after that cycle closed and were prosecuted by a second self-audit cycle: four rounds across all six registers, planted controls caught three of three in every round at SELF grade, seven findings repaired or scoped, terminal verdict HALT-DIVERGENT, computed from earned corrections rising across the last three rounds, with the single-substrate aperture open. A third self-audit cycle then prosecuted an external review supplied by the architect, finding by finding: four rounds across all six registers, planted controls caught three of three in every round at SELF grade, nineteen findings (eleven earned or scoped repairs, two refused on the text, two apertures), terminal verdict HALT-STALE. It graded every result K, A, or C, bound every equivalence to its imported inputs and Postulate M to its seed, and reduced unitarity to its scalar sense. The title is held as locked by the architect's ruling; the abstract and Section 8 state what is completed. Independent recompilation of the Lean file remains the open witness. Section 6.12 and Part XIV were added after the third cycle closed and have not yet passed an audit cycle of their own.

Scope. The bridge, the monism and anchored witnesses, the de Bruijn witness and its arrow, the cone with its halt, the equivalence of Postulate M with the line property, and the demand correctly addressed with the terminal mark at the foundation, and the logical form of the hypothesis (Theorems 70 to 73) are kernel theorems. The unitarity of the modes on the line and stability as the line property are kernel theorems on the lattice stage and hold in general by the identity (6.2a). The analytic inputs are cited. The sign on the zeros is not a clause of any proof here.

APPENDIX A: RA_LI_BRIDGE.LEAN

SHA-256 f2d0f4e33e82ad5db4afa6f4d12cfb93619354bec0ffb32efac8f7be3ef18f03, Lean 4.19.0, core only, 1058 lines.

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1 /-
2 RA_Li_Bridge.lean · the bridge from RA's positivity to Li positivity on  $\zeta$ , typed.
3 Nothing here is an axiom. RA, Li's criterion, and the bridge enter as hypotheses,
4 so #print axioms shows exactly what the kernel used.  $\Delta M = 0$ .
5 -/
6
7 namespace RALi
8
9 /-- The existents and their energies, as in ROOT, but as parameters. -/
10 structure Substrate where
11   U : Type
12    $\Delta E : U \rightarrow \text{Int}$ 
13
14 /-- RA as a hypothesis on a substrate: to exist is to actuate. -/
15 def RA (S : Substrate) : Prop :=  $\forall x : S.U, 0 < S.\Delta E\ x$ 
16
17 /-- An L-function reduced to what Li's criterion reads: its RH, and the
18   sign bit of each Li coefficient  $\lambda_n$  ( $n \geq 1$ ). -/
19 structure LiData where
20   RH : Prop
21   nonneg :  $\text{Nat} \rightarrow \text{Prop}$ 
22
23 /-- Li 1997 (Bombieri-Lagarias 1999 for the general case), cited, not proved:
24   RH  $\leftrightarrow$  every  $\lambda_n \geq 0$ . Carried as a hypothesis on the data. -/
25 def LiCriterion (L : LiData) : Prop :=  $L.RH \leftrightarrow \forall n, 1 \leq n \rightarrow L.\text{nonneg}\ n$ 
26
27 /-- THE BRIDGE. One existent per Li step, and a reading that turns its
28   positive energy into the sign of  $\lambda_n$ . The whole content lives in 'read'. -/
29 structure Bridge (S : Substrate) (L : LiData) where
30    $\phi : \text{Nat} \rightarrow S.U$ 
31   read :  $\forall n, 1 \leq n \rightarrow 0 < S.\Delta E\ (\phi\ n) \rightarrow L.\text{nonneg}\ n$ 
32
33 /-- 1. The bridge closes: RA, a bridge, and Li's criterion give RH. -/
34 theorem bridge_yields_RH (S : Substrate) (L : LiData)
35   (hRA : RA S) (hLi : LiCriterion L) (B : Bridge S L) : L.RH :=
36   hLi.mpr (fun n hn => B.read n hn (hRA (B. $\phi$  n)))
37
38 /-- 2. The bridge is keyed: RA alone never supplies it. A substrate where RA
39   holds, paired with data carrying one negative  $\lambda$ , admits no bridge. -/
40 def oneEnergy : Substrate := (Unit, fun _ => 1)
41 def badData : LiData := (False, fun n => n  $\neq$  5)
42
43 theorem ra_holds_in_model : RA oneEnergy := fun _ => show (0 : Int) < 1 by decide
44
45 theorem bridge_is_keyed : RA oneEnergy  $\wedge \neg \text{Nonempty} (\text{Bridge oneEnergy badData}) :=$  by
46   refine (ra_holds_in_model, fun (B) => ?_)
47   exact B.read 5 (by decide) (ra_holds_in_model (B. $\phi$  5)) rfl
48
49 /-- 3. No uniform bridge. If the reading does not consume the L-function,
50   it forces positivity on every member of the family. One member with a
51   negative  $\lambda$  (the Eisenstein L-function  $\zeta(s)\zeta(s-k+1)$ : Euler product and
52   functional equation, zeros off its centre, so some  $\lambda_n < 0$  by
53   Bombieri-Lagarias) kills every uniform bridge. -/
54 structure UniformBridge (S : Substrate) (F : Type) (L : F  $\rightarrow$  LiData) where
55    $\phi : \text{Nat} \rightarrow S.U$ 
56   read :  $\forall f\ n, 1 \leq n \rightarrow 0 < S.\Delta E\ (\phi\ n) \rightarrow (L\ f).\text{nonneg}\ n$ 
57
58 theorem no_uniform_bridge (S : Substrate) (F : Type) (L : F  $\rightarrow$  LiData)
59   (hRA : RA S) (hctl :  $\exists f\ n, 1 \leq n \wedge \neg (L\ f).\text{nonneg}\ n$ ) :
60    $\neg \text{Nonempty} (\text{UniformBridge S F L}) :=$  by
61   intro (B)
62   obtain (f, n, hn, hneg) := hctl
63   exact hneg (B.read f n hn (hRA (B. $\phi$  n)))
64
65 /-- 4. The type repair. The arrow supplies one bit; Li turns RH's vanishing
66   into one sign bit per step. Given decidable signs, the Li family is a
67    $\text{Nat} \rightarrow \text{Bool}$ , the shape the deed can supply, and RH is its constancy. -/
68 theorem li_is_a_bit_stream (L : LiData) (hLi : LiCriterion L)
69   [dec :  $\forall n, \text{Decidable} (L.\text{nonneg}\ n)$ ] :
70    $L.RH \leftrightarrow \forall n, 1 \leq n \rightarrow \text{decide} (L.\text{nonneg}\ n) = \text{true} :=$  by
71   refine hLi.trans ?_
72   exact (fun h n hn => decide_eq_true (h n hn), fun h n hn => of_decide_eq_true (h n hn))
73
74 end RALi
75
76 #print axioms RALi.bridge_yields_RH
77 #print axioms RALi.bridge_is_keyed
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78 #print axioms RALi.no_uniform_bridge
79 #print axioms RALi.li_is_a_bit_stream
80
81 /-! ## PART II · Does  $[\Xi_0]$  retire on the united register? -/
82 namespace RALi
83
84 inductive Grade | premise | structural | theorem deriving DecidableEq, Repr
85 def Grade.rank : Grade → Nat | .premise => 0 | .structural => 1 | .theorem => 2
86 def Grade.weakest (a b : Grade) : Grade := if a.rank ≤ b.rank then a else b
87
88 inductive Verdict | seal (g : Grade) | xi0 deriving DecidableEq, Repr
89
90 /- The united register. Unity is the claim that the formal reading and the
91 kinetic reading are one; on  $\zeta$  it is stated as the supply of the bridge,
92 carried at the grade of whatever supplies it. -/
93 structure Unity (S : Substrate) (L : LiData) where
94   bridge : Bridge S L
95   grade : Grade
96
97 /- The emitter on the formal string: with unity supplied, the verdict is a
98 seal at the weakest grade on the chain; without it,  $[\Xi_0]$ . -/
99 def emit {S : Substrate} {L : LiData} : Option (Unity S L) → Verdict
100 | some u => .seal (Grade.weakest u.grade .theorem)
101 | none => .xi0
102
103 /- 5.  $[\Xi_0]$  retires exactly when unity is supplied, and the seal it becomes
104 is sound: it carries RH. -/
105 theorem xi0_retires_iff_unity (S : Substrate) (L : LiData) (u : Option (Unity S L)) :
106   emit u ≠ .xi0 ↔ u.isSome := by
107   cases u <|> simp [emit]
108
109 theorem retired_seal_is_sound (S : Substrate) (L : LiData)
110   (hRA : RA S) (hLi : LiCriterion L) (u : Unity S L) : L.RH :=
111   bridge_yields_RH S L hRA hLi u.bridge
112
113 /- 6. The grade law: the retired verdict is never stronger than its unity
114 posit. A premise-grade unity yields a premise-grade seal. -/
115 theorem retired_grade_capped {S : Substrate} {L : LiData} (u : Unity S L) :
116   emit (some u) = .seal (Grade.weakest u.grade .theorem) ∧
117   (Grade.weakest u.grade .theorem).rank ≤ u.grade.rank := by
118   refine {rfl, ?_}
119   cases h : u.grade <|> decide
120
121 /- 7. Unity is not free: RA does not supply it (from theorem 2). -/
122 theorem unity_is_keyed : RA oneEnergy ∧ ¬ Nonempty (Unity oneEnergy badData) :=
123   (ra_holds_in_model, fun (u) => bridge_is_keyed.2 {u.bridge})
124
125 /- 8. The Ghost. A reading that deletes the arrow returns one verdict for a
126 claim and its negation: it cannot be a verdict on either. -/
127 def arrowDeleted (P : Prop) : Verdict := .xi0
128 theorem ghost (P : Prop) : arrowDeleted P = arrowDeleted (¬ P) := rfl
129
130 end RALi
131
132 #print axioms RALi.xi0_retires_iff_unity
133 #print axioms RALi.retired_seal_is_sound
134 #print axioms RALi.retired_grade_capped
135 #print axioms RALi.unity_is_keyed
136 #print axioms RALi.ghost
137
138 /-! ## PART III · The try. Which Codex instruments can supply `Bridge.read` for  $\zeta$ ? -/
139 namespace RALi
140
141 /- 9. The uniformity trap. A family of bridges that picks its existents
142 without reading which L-function it serves is a uniform bridge. Every
143 instrument carried at  $\Delta M = 0$  (the eliminator, the multi-recursion
144 collapse, the one-bit species, the bare arrow) chooses its existents
145 that way, so it lands here. -/
146 theorem common_phi_is_uniform (S : Substrate) (F : Type) (L : F → LiData)
147   ( $\phi_0$  : Nat → S.U) (B : ∀ f, Bridge S (L f)) (h $\phi$  : ∀ f, (B f). $\phi$  =  $\phi_0$ ) :
148   Nonempty (UniformBridge S F L) :=
149   (( $\phi_0$ , fun f n hn h => (B f).read n hn (by rw [h $\phi$  f]; exact h)))
150
151 /- 10. Hence dead: with RA and one family member carrying a negative  $\lambda$ ,
152 no  $\zeta$ -blind choice of existents can serve every member. -/
153 theorem blind_constructions_die (S : Substrate) (F : Type) (L : F → LiData)
154   (hRA : RA S) (hctl : ∃ f n, 1 ≤ n ∧ ¬ (L f).nonneg n)
155   ( $\phi_0$  : Nat → S.U) : ¬ ∃ B : (∀ f, Bridge S (L f)), ∀ f, (B f). $\phi$  =  $\phi_0$  :=
156   fun (B, h $\phi$ ) => no_uniform_bridge S F L hRA hctl (common_phi_is_uniform S F L  $\phi_0$  B h $\phi$ )
157
158 /- 11. What positivity on the Euler product does reach. Mertens' identity
159  $3 + 4 \cos \theta + \cos 2\theta = 2(1 + \cos \theta)^2 \geq 0$ , with  $\cos 2\theta = 2c^2 - 1$ , is the
160 positivity that yields  $\zeta(1 + it) \neq 0$ : the wall at  $\text{Re} = 1$ . Checked here on

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161 integer c as the polynomial identity it is. -/
162 theorem mertens_identity (c : Int) : 3 + 4 * c + (2 * c * c - 1) = 2 * ((1 + c) * (1 + c)) := by
163   simp only [Int.add_mul, Int.mul_add, Int.one_mul, Int.mul_one, Int.mul_assoc]
164   omega
165
166 theorem mertens_nonneg (c : Int) : 0 ≤ 3 + 4 * c + (2 * c * c - 1) := by
167   rw [mertens_identity]
168   refine Int.mul_nonneg (by decide) ?_
169   rcases Int.le_total 0 (1 + c) with h | h
170   · exact Int.mul_nonneg h h
171   · have h' : 0 ≤ -(1 + c) := by omega
172     have := Int.mul_nonneg h' h'
173     rwa [Int.neg_mul_neg] at this
174
175 end RALi
176
177 #print axioms RALi.common_phi_is_uniform
178 #print axioms RALi.blind_constructions_die
179 #print axioms RALi.mertens_nonneg
180
181 /-! ## PART IV · Time, made exact. The Li modes on the Bridge plane.
182 A zero  $p = h/2 + i t$  sits at  $(h, t)$  in half-units, the Bridge plane, fold  $(h, t) \mapsto (2-h, t)$ .
183 Li's coefficient is  $\lambda_n = \sum_p [1 - z p^n]$  with the mode  $z p = 1 - 1/p = (p-1)/p$ , so step
184  $n \mapsto n+1$  multiplies each mode by  $z p$ .  $|z p|^2 = |p-1|^2/|p|^2$ , scaled by 4:
185  $N1 = (h-2)^2 + 4t^2$ ,  $N0 = h^2 + 4t^2$ .
186 A mode is unitary (perpetual, neither growing nor dying) iff  $N1 = N0$ ; it grows iff  $N1 > N0$ . -/
187 namespace RALi
188
189 def N1 (h t : Int) : Int := (h - 2) * (h - 2) + 4 * (t * t)
190 def N0 (h t : Int) : Int := h * h + 4 * (t * t)
191
192 theorem N1_sub_N0 (h t : Int) : N1 h t - N0 h t = 4 - 4 * h := by
193   simp only [N1, N0, Int.sub_mul, Int.mul_sub]
194   omega
195
196 /- 12. A mode is unitary exactly on the line: perpetuity in time is the fixed locus. -/
197 theorem unitary_iff_on_line (h t : Int) : N1 h t = N0 h t ↔ h = 1 := by
198   have := N1_sub_N0 h t
199   exact (fun e => by omega, fun e => by omega)
200
201 /- 13. Left of the line the mode grows; right of it the mode dies. -/
202 theorem grows_left (h t : Int) (hl : h < 1) : N0 h t < N1 h t := by
203   have := N1_sub_N0 h t; omega
204 theorem dies_right (h t : Int) (hr : 1 < h) : N1 h t < N0 h t := by
205   have := N1_sub_N0 h t; omega
206
207 /- 14. The fold forbids a quiet exit: any off-line zero, together with its mirror under
208 the functional equation, carries a growing mode. -/
209 theorem off_line_forces_growth (h t : Int) (hoff : h ≠ 1) :
210   N0 h t < N1 h t ∨ N0 (2 - h) t < N1 (2 - h) t := by
211   have a := N1_sub_N0 h t; have b := N1_sub_N0 (2 - h) t; omega
212
213 /- 15. PERPETUAL STABILITY IS THE LINE PROPERTY. For any fold-invariant zero set on the
214 plane: no mode grows under the time step iff every zero lies on  $h = 1$ . -/
215 theorem stability_iff_line (Z : Int × Int → Prop)
216   (hinv : ∀ p, Z p → Z (2 - p.1, p.2)) :
217   (∀ p, Z p → N1 p.1 p.2 ≤ N0 p.1 p.2) ↔ (∀ p, Z p → p.1 = 1) := by
218   constructor
219   · intro hs p hp
220     have a := hs p hp
221     have b := hs (2 - p.1, p.2) (hinv p hp)
222     have c := N1_sub_N0 p.1 p.2
223     have d := N1_sub_N0 (2 - p.1) p.2
224     simp only at b d
225     omega
226   · intro hl p hp
227     have := hl p hp
228     have c := N1_sub_N0 p.1 p.2
229     omega
230
231 end RALi
232
233 #print axioms RALi.unitary_iff_on_line
234 #print axioms RALi.off_line_forces_growth
235 #print axioms RALi.stability_iff_line
236
237 /-! ## PART V · The monism rule, hardened. "What happens once happens again." -/
238 namespace RALi
239
240 /- 16. One repeating one: the rule, with its step, is induction. -/
241 theorem one_repeats_one (P : Nat → Prop) (h0 : P 0) (step : ∀ n, P n → P (n + 1)) :
242   ∀ n, P n := by
243   intro n; induction n with

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244 | zero => exact h0
245 | succ k ih => exact step k ih
246
247 /-- 17. The step is the whole content: the universal claim is exactly the base plus the step. -/
248 theorem the_step_is_the_claim (P : Nat → Prop) :
249   (∀ n, P n) ↔ (P 0 ∧ ∀ n, P n → P (n + 1)) :=
250   (fun h => (h 0, fun n _ => h (n + 1)), fun (h0, s) => one_repeats_one P h0 s)
251
252 /-- 18. The monism posit: one principle means the property does not vary with the index.
253 Under it, one instance forces all. -/
254 theorem monism_posit_forces (P : Nat → Prop) (uniform : ∀ n m, P n ↔ P m) (h0 : P 0) :
255   ∀ n, P n := fun n => (uniform 0 n).mp h0
256
257 /-- 19. Finite confirmation never forces. For every height N there is a property true on
258 every index below N and false beyond: Skewes, Mertens, Pólya are this shape. -/
259 theorem finite_never_forces (N : Nat) :
260   ∃ P : Nat → Prop, (∀ n, n < N → P n) ∧ ¬ ∀ n, P n :=
261   (fun n => n < N, fun _ h => h, fun h => Nat.lt_irrefl N (h N))
262
263 /-- 20. Monism is not a law of the fold. A fold-invariant zero set can hold one zero on the
264 line and a mirror pair off it: "on the line once" does not recur by symmetry alone.
265 This is the Davenport-Heilbronn shape on the Bridge plane. -/
266 def mixedZ (p : Int × Int) : Prop := p = (1, 0) ∨ p = (0, 5) ∨ p = (2, 5)
267
268 theorem mixed_is_fold_invariant : ∀ p, mixedZ p → mixedZ (2 - p.1, p.2) := by
269   intro p hp
270   rcases hp with h | h | h <=> subst h
271   · exact 0r.inl rfl
272   · exact 0r.inr (0r.inr rfl)
273   · exact 0r.inr (0r.inl rfl)
274
275 theorem mixed_recurrence_fails :
276   mixedZ (1, 0) ∧ mixedZ (0, 5) ∧ ¬ (∀ p, mixedZ p → p.1 = 1) :=
277   (0r.inl rfl, 0r.inr (0r.inl rfl), fun h => by
278     have := h (0, 5) (0r.inr (0r.inl rfl)); cases this)
279
280 end RALi
281
282 #print axioms RALi.one_repeats_one
283 #print axioms RALi.the_step_is_the_claim
284 #print axioms RALi.monism_posit_forces
285 #print axioms RALi.finite_never_forces
286 #print axioms RALi.mixed_is_fold_invariant
287 #print axioms RALi.mixed_recurrence_fails
288
289 /-! ## PART VI · Actualism. Only measured zeros are actual. -/
290 namespace RALi
291
292 /-- A zero set on a carrier, a line predicate, and the time at which each zero is measured
293 (none if never). The actual zeros at time T are those measured by T. -/
294 structure Actualized (α : Type) where
295   Z      : α → Prop
296   onLine : α → Prop
297   measured : α → Option Nat
298
299 def RHfull {α : Type} (A : Actualized α) : Prop := ∀ z, A.Z z → A.onLine z
300 def RHactual {α : Type} (A : Actualized α) : Prop :=
301   ∀ z, A.Z z → A.measured z ≠ none → A.onLine z
302
303 /-- 21. The full claim yields the actual one. -/
304 theorem full_gives_actual {α : Type} (A : Actualized α) : RHfull A → RHactual A :=
305   fun h z hz _ => h z hz
306
307 /-- 22. The actual claim does not yield the full one: one never-measured zero off the line. -/
308 def unreached : Actualized Bool := {fun _ => True, fun b => b = true, fun b => if b then some 0 else none}
309
310 theorem actual_not_full : RHactual unreached ∧ ¬ RHfull unreached := by
311   refine {fun z _ hm => ?_, fun h => by cases h false trivial}
312   cases z
313   · exact absurd rfl hm
314   · rfl
315
316 /-- 23. Given sufficient time, every zero is measured, and then the actual claim over all
317 time is exactly the full claim: actualism re-routes to the same step and does not bypass it. -/
318 theorem sufficient_time {α : Type} (A : Actualized α) (reach : ∀ z, A.Z z → A.measured z ≠ none) :
319   RHactual A ↔ RHfull A :=
320   (fun h z hz => h z hz (reach z hz), full_gives_actual A)
321
322 end RALi
323
324 #print axioms RALi.full_gives_actual
325 #print axioms RALi.actual_not_full
326 #print axioms RALi.sufficient_time

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327
328 /-! ## PART VII · The constructed witness, reconstructed from Residual Monism.
329 The paper's witness is built from RA and read through RAF and RAM. Here the witness is
330 built from the Codex connector directly: one substrate, one involution serving both routes
331 ( $\sigma = \sigma'$ ), and one principle that does not vary with the index. The index is time. -/
332 namespace RALi
333
334 structure Q4 where
335   r : Int
336   i : Int
337   j : Int
338   k : Int
339   deriving DecidableEq, Repr
340
341 /- The geometric route's involution and the formal route's involution, written apart. -/
342 def sigmaGeo (q : Q4) : Q4 := (q.r, -q.i, -q.j, -q.k)
343 def sigmaForm (q : Q4) : Q4 := (q.r, -q.i, -q.j, -q.k)
344
345 /- THE MONISM WITNESS. One involution, one principle, one seed. -/
346 structure MonismWitness (P : Nat → Prop) : Prop where
347   one_involution : ∀ q, sigmaGeo q = sigmaForm q
348   uniform       : ∀ n m, P n ↔ P m
349   seed          : P 0
350
351 /- 24. The seat field is constructed, not posited:  $\sigma = \sigma'$  holds by rfl. -/
352 theorem one_involution_constructed : ∀ q, sigmaGeo q = sigmaForm q := fun _ => rfl
353
354 /- 25. The timeless reading and the timed reading coincide under one principle:
355 the claim over all time is the claim at one instant. Deleting time loses nothing. -/
356 theorem timeless_equals_timed (P : Nat → Prop) (u : ∀ n m, P n ↔ P m) :
357   (∀ n, P n) ↔ P 0 :=
358   (fun h => h 0, fun h n => (u 0 n).mp h)
359
360 /- 26. The witness closes the record, past and future. -/
361 theorem monism_closes (P : Nat → Prop) (w : MonismWitness P) : ∀ n, P n :=
362   (timeless_equals_timed P w.uniform).mpr w.seed
363
364 /- 27. The witness is exactly the claim: a monism witness for P exists iff P holds at
365 every index. Its value field carries the whole of the claim, as theorem 17 required. -/
366 theorem monism_witness_is_the_claim (P : Nat → Prop) :
367   MonismWitness P ↔ ∀ n, P n :=
368   (monism_closes P, fun h => (one_involution_constructed,
369     fun n m => (fun _ => h m, fun _ => h n), h 0))
370
371 /- 28. Constructed at the model, as the paper constructs RA at one point. -/
372 theorem constructed_monism : MonismWitness (fun _ => True) :=
373   (one_involution_constructed, fun _ _ => Iff.rfl, trivial)
374
375 /- 29. Unlike the RA witness, the monism witness is not silent on counter-models:
376 it cannot be constructed where the property breaks, so it decides by containing. -/
377 theorem monism_absent_on_breaks : ¬ MonismWitness (fun n => n = 0) :=
378   fun w => absurd ((w.uniform 0 1).mp rfl) (by decide)
379
380 /- 30. Run through the RA-Li bridge. Replace the bridge by the monism witness on Li's
381 sign stream: the witness, with Li's criterion, gives RH. -/
382 theorem monism_yields_RH (L : LiData) (hLi : LiCriterion L)
383   (w : MonismWitness (fun n => L.nonneg (n + 1))) : L.RH :=
384   hLi.mpr (fun n hn => by
385     obtain (k, rfl) : ∃ k, n = k + 1 := (n - 1, by omega)
386     exact monism_closes _ w k)
387
388 /- 31. And the price, stated: on  $\zeta$  the witness exists iff every Li sign is nonnegative,
389 which by Li is RH. Reconstruction moves the owed bit into `uniform`; it does not remove it. -/
390 theorem monism_on_li_is_RH (L : LiData) (hLi : LiCriterion L) :
391   MonismWitness (fun n => L.nonneg (n + 1)) ↔ L.RH := by
392   rw [monism_witness_is_the_claim]
393   refine (fun h => monism_yields_RH L hLi ((monism_witness_is_the_claim _).mpr h), fun h n => ?_)
394   exact (hLi.mp h) (n + 1) (by omega)
395
396 end RALi
397
398 #print axioms RALi.one_involution_constructed
399 #print axioms RALi.timeless_equals_timed
400 #print axioms RALi.monism_closes
401 #print axioms RALi.monism_witness_is_the_claim
402 #print axioms RALi.constructed_monism
403 #print axioms RALi.monism_absent_on_breaks
404 #print axioms RALi.monism_yields_RH
405 #print axioms RALi.monism_on_li_is_RH
406
407 /-! ## PART VIII · Re-anchoring the monism witness to mathematical standard.
408 Calibration case: Perelman. The standard form of "one principle, the same result again"
409 is not a free uniformity; it is one dynamics, one invariant, and a PROVED transport of the

```

```

410 invariant along the dynamics (induction on the orbit; Lyapunov monotonicity; Perelman's
411 W-entropy under Ricci flow). -/
412 namespace RALi
413
414 /-- 32. Monism as a schema over all properties is inconsistent: uniformity cannot be
415 posited for every P. It must be anchored to a principle, not spread over properties. -/
416 theorem global_monism_inconsistent : ¬ ∀ P : Nat → Prop, ∀ n m, P n ↔ P m :=
417   fun h => absurd ((h (fun n => n = 0) 0 1).mp rfl) (by decide)
418
419 def iter {S : Type} (f : S → S) : Nat → S → S
420 | 0, s => s
421 | n + 1, s => f (iter f n s)
422
423 /-- THE ANCHORED MONISM WITNESS. One substrate S, one principle f (the dynamics),
424 one invariant I, the transport of I along f, and a seed. -/
425 structure AnchoredMonism (S : Type) where
426   f      : S → S
427   I      : S → Prop
428   transport : ∀ s, I s → I (f s)
429   s0     : S
430   seed   : I s0
431
432 /-- 33. The anchored witness closes its whole orbit, past and future. -/
433 theorem anchored_closes {S : Type} (A : AnchoredMonism S) : ∀ n, A.I (iter A.f n A.s0) := by
434   intro n; induction n with
435   | zero => exact A.seed
436   | succ k ih => exact A.transport _ ih
437
438 /-- 34. The free monism witness is DERIVED from the anchored one: uniformity is no longer
439 posited, it is manufactured by the transport. -/
440 theorem anchored_derives_monism {S : Type} (A : AnchoredMonism S) :
441   MonismWitness (fun n => A.I (iter A.f n A.s0)) :=
442   (one_involution_constructed,
443     fun n m => (fun _ => anchored_closes A m, fun _ => anchored_closes A n),
444     A.seed)
445
446 /-- 35. THE PERELMAN SHAPE. A quantity monotone along one flow, here a discrete flow on its
447 time index, is an anchored monism witness for the property "never below its start." Perelman's
448 W-entropy along Ricci flow has this shape; it is a structural analogue and is not formalized. -/
449 def lyapunovWitness (W : Nat → Int) (mono : ∀ n, W n ≤ W (n + 1)) : AnchoredMonism Nat :=
450   (Nat.succ, fun n => W 0 ≤ W n, fun n h => Int.le_trans h (mono n), 0, Int.le_refl _)
451
452 theorem lyapunov_is_monism (W : Nat → Int) (mono : ∀ n, W n ≤ W (n + 1)) :
453   MonismWitness (fun n => W 0 ≤ W (iter Nat.succ n 0)) :=
454   anchored_derives_monism (lyapunovWitness W mono)
455
456 /-- 36. The calibration's grade law: the closure is as strong as its transport. -/
457 structure GradedAnchor (S : Type) where
458   A      : AnchoredMonism S
459   tgrade : Grade
460
461 def closureGrade {S : Type} (G : GradedAnchor S) : Grade := Grade.weakest G.tgrade .theorem
462
463 theorem closure_grade_is_transport_grade {S : Type} (G : GradedAnchor S) :
464   closureGrade G = G.tgrade := by
465   unfold closureGrade Grade.weakest; cases G.tgrade <=> rfl
466
467 /-- 37. The RH instance. On Li's sign stream with the time step as the principle, an
468 anchored witness exists iff RH. The transport is `nonneg (n+1) → nonneg (n+2)`: the
469 ζ-analogue of Perelman's monotonicity formula, not yet proved by anyone. -/
470 theorem rh_anchor_is_the_claim (L : LiData) (hLi : LiCriterion L) :
471   (∃ A : AnchoredMonism Nat, A.f = Nat.succ ∧ A.s0 = 0 ∧
472     ∀ n, A.I n ↔ L.nonneg (n + 1)) ↔ L.RH := by
473   constructor
474   · rintro (A, hf, hs, hI)
475     apply (monism_on_li_is_RH L hLi).mp
476     refine (monism_witness_is_the_claim _).mpr (fun n => ?_)
477     have key : ∀ n, iter A.f n A.s0 = n := by
478       intro n; induction n with
479       | zero => exact hs
480       | succ k ih => show A.f (iter A.f k A.s0) = k + 1; rw [ih, hf]
481     have := anchored_closes A n
482     rw [key n] at this
483     exact (hI n).mp this
484   · intro h
485     have all := (monism_witness_is_the_claim _).mp ((monism_on_li_is_RH L hLi).mpr h)
486     exact ((Nat.succ, fun n => L.nonneg (n + 1), fun n _ => all (n + 1), 0, all 0),
487       rfl, rfl, fun _ => Iff.rfl)
488
489 end RALi
490
491 #print axioms RALi.global_monism_inconsistent
492 #print axioms RALi.anchored_closes

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493 #print axioms RALi.anchored_derives_monism
494 #print axioms RALi.lyapunov_is_monism
495 #print axioms RALi.closure_grade_is_transport_grade
496 #print axioms RALi.rh_anchor_is_the_claim
497
498 /-! ## PART IX ·  $\zeta$ 's time has a proved monotonicity formula, and it runs the wrong way.
499 The de Bruijn-Newman heat flow  $H_t(z) = \int e^{\{t u^2\}} \Phi(u) \cos(z u) du$ ,  $\Phi$  built from  $\xi$ .
500 de Bruijn (1950): if every zero of  $H_0$  lies in  $|\operatorname{Im} z| \leq \Delta$ , every zero of  $H_t$  lies in
501  $|\operatorname{Im} z| \leq \sqrt{\max(\Delta^2 - 2t, 0)}$ . RH is "all zeros of  $H_0$  real".  $\Lambda$  is the least  $t$  with all zeros
502 of  $H_t$  real; Newman (1976) defined it, Rodgers-Tao (2020) proved  $\Lambda \geq 0$ , Polymath 15 (2019)
503 proved  $\Lambda \leq 0.22$ . So  $\text{RH} \leftrightarrow \Lambda = 0$ . Modelled here on the strip bound, widths in  $\text{Nat}$ , half-steps
504 of time; a toy of the bound, not of the flow. -/
505 namespace RALi
506
507 /- Squared strip width after  $t$  time steps:  $\max(d_2 - 2t, 0)$ . -/
508 def flow (d2 t : Nat) : Nat := d2 - 2 * t
509 def realAt (d2 t : Nat) : Prop := flow d2 t = 0
510
511 /- 38. THE MONOTONICITY FORMULA. The width never grows along  $\zeta$ 's time. -/
512 theorem flow_monotone (d2 t : Nat) : flow d2 (t + 1) ≤ flow d2 t := by
513   unfold flow; omega
514
515 /- 39. Its transport, proved: once every zero is real, every zero stays real. This is
516 de Bruijn's theorem in the toy, the  $\zeta$ -analogue of Perelman's monotonicity, at theorem grade. -/
517 theorem reality_transported (d2 t : Nat) (h : realAt d2 t) : realAt d2 (t + 1) := by
518   unfold realAt flow at *; omega
519
520 /-  $\Lambda$  in the toy: the first time the strip closes. -/
521 def Lam (d2 : Nat) : Nat := (d2 + 1) / 2
522
523 theorem real_at_Lam (d2 : Nat) : realAt d2 (Lam d2) := by unfold realAt flow Lam; omega
524
525 /- 40. In the toy, the strip is closed at time zero iff  $\Lambda = 0$ . For  $\zeta$ ,  $\text{RH} \leftrightarrow \Lambda = 0$  is the cited
526 result of Newman with Rodgers and Tao and enters the cone as ' $\text{hLam}$ '. -/
527 theorem rh_iff_lambda_zero (d2 : Nat) : realAt d2 0  $\leftrightarrow$  Lam d2 = 0 := by
528   unfold realAt flow Lam
529   exact (fun h => by omega, fun h => by omega)
530
531 /- 41. The anchored monism witness exists for  $\zeta$ 's time, transport proved, seeded at  $\Lambda$ .
532 It closes the whole future of the flow from  $\Lambda$  on. -/
533 def deBruijnWitness (d2 : Nat) : AnchoredMonism Nat :=
534   (Nat.succ, realAt d2, fun t h => reality_transported d2 t h, Lam d2, real_at_Lam d2)
535
536 theorem iter_succ (n s : Nat) : iter Nat.succ n s = s + n := by
537   induction n with
538   | zero => rfl
539   | succ k ih => show Nat.succ (iter Nat.succ k s) = s + (k + 1); rw [ih]; omega
540
541 theorem future_closed (d2 n : Nat) : realAt d2 (Lam d2 + n) := by
542   have := anchored_closes (deBruijnWitness d2) n
543   rwa [show (deBruijnWitness d2).f = Nat.succ from rfl, iter_succ] at this
544
545 /- 42. THE RECORD FORGETS. After one step the record of the width-zero state and the
546 width-one state coincide: the forward flow is even in the RH bit, so no readout of the
547 flowed record decides RH. This is the fTOE wall, T1, executed on  $\zeta$ 's own time. -/
548 theorem flowed_record_forgets (T : Nat) (hT : 1 ≤ T) :
549    $\neg \exists g : \text{Nat} \rightarrow \text{Bool}, \forall d_2, g (\text{flow } d_2 T) = \text{decide } (d_2 = 0) :=$  by
550   rintro (g, hg)
551   have a := hg 0; have b := hg 1
552   have e : flow 1 T = flow 0 T := by unfold flow; omega
553   rw [e, a] at b
554   exact absurd b (by decide)
555
556 /- 43. The upstream direction is not a transport. Going backward from a real record,
557 both answers are admissible: the preimage of "real at T" holds the RH state and a non-RH
558 state. The forward arrow proves; the backward arrow must be supplied. -/
559 theorem upstream_is_not_forced (T : Nat) (hT : 1 ≤ T) :
560   realAt 0 T  $\wedge$  realAt 1 T  $\wedge$  realAt 0 0  $\wedge$   $\neg$  realAt 1 0 := by
561   unfold realAt flow
562   exact (by omega, by omega, rfl, fun h => by omega)
563
564 end RALi
565
566 #print axioms RALi.flow_monotone
567 #print axioms RALi.reality_transported
568 #print axioms RALi.rh_iff_lambda_zero
569 #print axioms RALi.future_closed
570 #print axioms RALi.flowed_record_forgets
571 #print axioms RALi.upstream_is_not_forced
572
573 /-! ## PART X · Halted Uniduction with the category-gap eliminator. The regress, closed.
574 Every "missing piece" of Parts I-IX is a register's reading of one seat. Exhibit each as a
575 leg of one cone with apex RH; then the hunt has one gap, not a sequence of them. -/

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```

576 namespace RALi
577
578 structure Hunt where
579   S      : Substrate
580   L      : LiData
581   u0     : S.U
582   hRA    : RA S
583   hLi    : LiCriterion L
584   lamZero : Prop      -- the upstream reading:  $\Lambda = 0$ 
585   hLam   : L.RH  $\leftrightarrow$  lamZero      -- Newman's definition with Rodgers-Tao, cited
586
587 /-- The legs. Each register's reading of the owed seat, proved equal to the apex. -/
588 theorem leg_bridge (H : Hunt) : Nonempty (Bridge H.S H.L)  $\leftrightarrow$  H.L.RH :=
589   (fun (B) => bridge_yields_RH H.S H.L H.hRA H.hLi B,
590    fun h => ((fun _ => H.u0, fun n hn _ => (H.hLi.mp h) n hn)))
591
592 theorem leg_unity (H : Hunt) : Nonempty (Unity H.S H.L)  $\leftrightarrow$  H.L.RH :=
593   (fun (u) => retired_seal_is_sound H.S H.L H.hRA H.hLi u,
594    fun h => let (B) := (leg_bridge H).mpr h; ((B, .premise)))
595
596 theorem leg_monism (H : Hunt) : MonismWitness (fun n => H.L.nonneg (n + 1))  $\leftrightarrow$  H.L.RH :=
597   monism_on_li_is_RH H.L H.hLi
598
599 theorem leg_transport (H : Hunt) :
600   ( $\exists$  A : AnchoredMonism Nat, A.f = Nat.succ  $\wedge$  A.s0 = 0  $\wedge$   $\forall$  n, A.I n  $\leftrightarrow$  H.L.nonneg (n + 1))
601    $\leftrightarrow$  H.L.RH :=
602   rh_anchor_is_the_claim H.L H.hLi
603
604 theorem leg_upstream (H : Hunt) : H.lamZero  $\leftrightarrow$  H.L.RH := H.hLam.symm
605
606 /-- The five readings, as a diagram of propositions. -/
607 inductive Reg5 | bridge | unity | monism | transport | upstream deriving DecidableEq, Repr
608
609 def D5 (H : Hunt) : Reg5  $\rightarrow$  Prop
610 | .bridge    => Nonempty (Bridge H.S H.L)
611 | .unity     => Nonempty (Unity H.S H.L)
612 | .monism    => MonismWitness (fun n => H.L.nonneg (n + 1))
613 | .transport =>  $\exists$  A : AnchoredMonism Nat, A.f = Nat.succ  $\wedge$  A.s0 = 0  $\wedge$   $\forall$  n, A.I n  $\leftrightarrow$  H.L.nonneg (n + 1)
614 | .upstream  => H.lamZero
615
616 /-- 44. THE CONE. RH is the apex of a cone over all five readings, every leg an
617 equivalence: the category gaps between the hunt's registers are eliminated. -/
618 theorem hunt_cone (H : Hunt) :  $\forall$  r, D5 H r  $\leftrightarrow$  H.L.RH := by
619   intro r; cases r
620   · exact leg_bridge H
621   · exact leg_unity H
622   · exact leg_monism H
623   · exact leg_transport H
624   · exact leg_upstream H
625
626 /-- 45. ONE GAP. Any two readings are the same proposition. The quantity monotone against
627 the flow (upstream) IS the bridge's read, the unity posit, the monism field, the transport. -/
628 theorem one_gap (H : Hunt) (r s : Reg5) : D5 H r  $\leftrightarrow$  D5 H s :=
629   (hunt_cone H r).trans (hunt_cone H s).symm
630
631 /-- 46. THE HALT. Every proposition proved equivalent to the hypothesis lands on the same apex
632 as the five readings; which future propositions are equivalent is not decided here. -/
633 theorem regress_halts (H : Hunt) (P : Prop) (hP : P  $\leftrightarrow$  H.L.RH) (r : Reg5) : P  $\leftrightarrow$  D5 H r :=
634   hP.trans (hunt_cone H r).symm
635
636 /-- 47. Given a true proposition Q, proving Q  $\rightarrow$  RH is the same as proving RH: conditioning on a
637 true premise adds nothing (Theorem E of the prior paper, here on the hunt's apex). -/
638 theorem nothing_weaker (H : Hunt) (Q : Prop) (hq : Q) : (Q  $\rightarrow$  H.L.RH)  $\leftrightarrow$  H.L.RH :=
639   (fun f => f hq, fun h _ => h)
640
641 end RALi
642
643 #print axioms RALi.hunt_cone
644 #print axioms RALi.one_gap
645 #print axioms RALi.regress_halts
646 #print axioms RALi.nothing_weaker
647
648 /-! ## PART X, completed · a sufficient route closes every reading at once. -/
649 namespace RALi
650
651 /-- 48. Theorem 46 covers every reformulation equivalent to the apex. A route strictly
652 stronger than the apex (a specific operator, a hypothesis over a family) is not equivalent,
653 and it does not open a new gap either: any sufficient route supplies all five readings. -/
654 theorem sufficient_closes_all (H : Hunt) (P : Prop) (hP : P  $\rightarrow$  H.L.RH) (p : P) :
655    $\forall$  r, D5 H r :=
656   fun r => (hunt_cone H r).mpr (hP p)
657
658 end RALi

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742 same_locus      : SameLocus W
743 timeless        : Timeless W
744 interface        : P W 0          -- every zero present at the timeless interface is on the line
745
746 /-- 55. When it can be built, it proves the bound everywhere downstream. -/
747 theorem witness_proves_bound (W : World) (w : MonismTimeWitness W) : Bound W :=
748   timeless_with_seed_binds W w.same_locus w.timeless w.interface
749
750 /-- 55, completed. It can be built exactly when the bound already holds: its existence IS the line property. -/
751 theorem witness_iff_bound (W : World) (h : SameLocus W) :
752   Nonempty (MonismTimeWitness W) ↔ Bound W := by
753   refine (fun {w} => witness_proves_bound W w, fun b => ?_)
754   have l := (bound_iff_line W h).mp b
755   exact {(fun _ => rfl, h, fun t s => (fun _ p z => l s p z, fun _ p z => l t p z), fun p z => l 0 p z)}
756
757 /-- 56. Built at a model whose zeros are on the line, as the prior paper builds RA at one point. -/
758 def goodWorld : World := {fun _ p => onLine p, fun _ p => p = (1, 0)}
759 theorem constructed_at_model : Nonempty (MonismTimeWitness goodWorld) :=
760   (witness_iff_bound goodWorld (fun _ _ => Iff.rfl)).mpr (fun _ p z => by subst z; rfl)
761
762 /-- 57. Not buildable at the counter-world: the interface field fails there. -/
763 theorem not_constructible_off_line : ¬ Nonempty (MonismTimeWitness testWorld) :=
764   fun w => test_not_bound ((witness_iff_bound testWorld test_same_locus).mp w)
765
766 end TimeLocus
767
768 namespace TimeLocus
769
770 /-- Postulate M on an enumerated zero set. Stage k is when a zero is located; L t says every
771   zero located by stage t lies on the line; M says the truth of L t does not depend on t. -/
772 def Lstage {α : Type} (Z : α → Prop) (onL : α → Prop) (stage : α → Nat) (t : Nat) : Prop :=
773   ∀ z, Z z → stage z ≤ t → onL z
774 def PostulateM {α : Type} (Z : α → Prop) (onL : α → Prop) (stage : α → Nat) : Prop :=
775   ∀ t s, Lstage Z onL stage t ↔ Lstage Z onL stage s
776
777 /-- 49. POSTULATE M IS EQUIVALENT TO THE LINE PROPERTY, given the computed seed L 0 and an
778   exhaustive enumeration (every zero is located at some finite stage, built into 'stage'). -/
779 theorem postulateM_iff_line {α : Type} (Z : α → Prop) (onL : α → Prop) (stage : α → Nat)
780   (seed : Lstage Z onL stage 0) :
781   PostulateM Z onL stage ↔ ∀ z, Z z → onL z := by
782   constructor
783   · intro m z hz
784     exact ((m 0 (stage z)).mp seed) z hz (Nat.le_refl _)
785   · intro h t s
786     exact {fun _ z hz _ => h z hz, fun _ z hz _ => h z hz}
787
788 /-- 50. And Postulate M is not free: it holds in a world whose zeros are all on the line, and
789   it cannot hold with the seed in a world with a located off-line zero. -/
790 theorem postulateM_decides {α : Type} (Z : α → Prop) (onL : α → Prop) (stage : α → Nat)
791   (seed : Lstage Z onL stage 0) (z : α) (hz : Z z) (off : ¬ onL z) :
792   ¬ PostulateM Z onL stage :=
793   fun m => off ((postulateM_iff_line Z onL stage seed).mp m z hz)
794
795 end TimeLocus
796
797 #print axioms TimeLocus.bound_iff_line
798 #print axioms TimeLocus.time_does_not_bind
799 #print axioms TimeLocus.monism_holds_in_counter_world
800 #print axioms TimeLocus.timeless_with_seed_binds
801 #print axioms TimeLocus.witness_iff_bound
802 #print axioms TimeLocus.constructed_at_model
803 #print axioms TimeLocus.not_constructible_off_line
804 #print axioms TimeLocus.postulateM_iff_line
805 #print axioms TimeLocus.postulateM_decides
806
807 /-! ## PART XII · The demand correctly addressed, and [.] at the collapse foundation. -/
808 namespace DotRH
809
810 inductive Token where
811   | sealed | broken | opn | dot
812   deriving DecidableEq, Repr
813
814 def Token.inEconomy : Token → Bool
815   | .dot => false
816   | _   => true
817
818 /-- I · THE COLLAPSE FOUNDATION and its stop. -/
819 structure Collapse (V : Prop) where
820   Reg      : Type
821   reading  : Reg → Prop
822   leg      : ∀ r, reading r ↔ V
823
824 -- Theorem 58.

```

```

825 theorem self_referential_limit {V : Prop} (C : Collapse V) (r s : C.Reg) :
826   (C.reading r ↔ V) ∧ (C.reading r ↔ C.reading s) :=
827   (C.leg r, (C.leg r).trans (C.leg s).symm)
828
829 def emitAtFoundation {V : Prop} (C : Collapse V) : Token := .dot
830 -- Theorem 59.
831 theorem value_marked_dot {V : Prop} (C : Collapse V) : emitAtFoundation C = .dot := rfl
832 theorem dot_is_not_a_verdict :
833   Token.dot ≠ .sealed ∧ Token.dot ≠ .broken ∧ Token.dot ≠ .opn := by decide
834 theorem dot_outside_economy : Token.inEconomy .dot = false := rfl
835
836 def trivialCollapse (V : Prop) : Collapse V := (Unit, fun _ => V, fun _ => Iff.rfl)
837
838 /-- Inside the collapse register, a demand for a verdict is a ghost: the register returns
839   the same mark for the value and its negation. -/
840 -- Theorem 60.
841 theorem demand_is_ghost_in_register {V : Prop} (C : Collapse V) (C' : Collapse (¬ V)) :
842   emitAtFoundation C = emitAtFoundation C' := rfl
843
844 /-- II · THE SELF-GROUNDING ROOT. A root R is self-grounding when acts occur and every act of
845   adjudication, assent or denial, is itself an instance of R. Then a denial of R re-enacts R,
846   no external proof adds to it, and a demand for one is a ghost. This is the seat of the
847   analogy: a root that grounds itself cannot be proved from outside itself. -/
848 structure SelfGrounding (R : Prop) where
849   Act      : Type
850   anAct    : Act
851   instances : Act → R
852
853 -- Theorem 61.
854 theorem denial_reenacts_root {R : Prop} (G : SelfGrounding R) (denial : G.Act) : R :=
855   G.instances denial
856
857 -- Theorem 61, completed.
858 theorem external_proof_adds_nothing {R : Prop} (G : SelfGrounding R) (Q : Prop) :
859   (Q → R) ↔ R :=
860   (fun _ => G.instances G.anAct, fun r _ => r)
861
862 /-- The Root Axiom at the constructed domain is self-grounding: every act is a deed, and a
863   deed actuates. -/
864 def RA : Prop := (0 : Int) < 1
865 def raSelfGrounding : SelfGrounding RA := (Unit, (), fun _ => show (0 : Int) < 1 by decide)
866
867 /-- III · A HYPOTHESIS ABOUT AN OBJECT is not self-grounding. Its value is fixed by the object,
868   acts do not instance it, and a finite witness can refute it. -/
869 structure Frame where
870   S : Type
871   τ : S → S
872   Z : S → Prop
873
874 def LineProperty (X : Frame) : Prop := ∀ s, X.Z s → X.τ s = s
875
876 /-- A located zero off the line refutes the line property, constructively. -/
877 theorem refuted_by_witness (X : Frame) (s : X.S) (hz : X.Z s) (off : X.τ s ≠ s) :
878   ¬ LineProperty X :=
879   fun h => off (h s hz)
880
881 def twoPoint : Frame := (Bool, fun b => !b, fun _ => True)
882
883 /-- On a frame whose line property fails, no occurring act can instance it: the line
884   property of a frame is not self-grounding. -/
885 -- Theorem 62.
886 theorem line_not_self_grounding : ¬ Nonempty (SelfGrounding (LineProperty twoPoint)) :=
887   fun (G) => refuted_by_witness twoPoint true trivial (fun h => Bool.noConfusion h) (G.instances G.anAct)
888
889 /-- IV · THE HARDENED FOUNDATION, in toto. The collapse marks the value [.] and the demand is
890   a ghost inside that register; the self-grounding root cannot be proved from outside and
891   re-enacts under denial; a hypothesis about an object is witness-refutable and is not
892   self-grounding, so the self-grounding exemption does not transfer to it. -/
893 theorem foundation_hardenened :
894   (∀ (V : Prop) (C : Collapse V), emitAtFoundation C = .dot) ∧
895   Token.inEconomy .dot = false ∧
896   (∀ (V : Prop) (C : Collapse V) (C' : Collapse (¬ V)), emitAtFoundation C = emitAtFoundation C') ∧
897   (∀ Q : Prop, (Q → RA) ↔ RA) ∧
898   (∀ a : Unit, RA ∧ raSelfGrounding.instances a = raSelfGrounding.instances a) ∧
899   ¬ Nonempty (SelfGrounding (LineProperty twoPoint)) :=
900   (fun _ => rfl, rfl, fun _ _ => rfl, external_proof_adds_nothing raSelfGrounding,
901     fun a => (raSelfGrounding.instances a, rfl), line_not_self_grounding)
902
903 /-! V · THE DEMAND, CORRECTLY ADDRESSED.
904   "A proof of the Riemann Hypothesis cannot be demanded of the foundation. Every reading the
905   foundation supplies is equivalent to the hypothesis, so any derivation from those readings
906   alone would presuppose what it derives. The demand is well-posed when addressed to the
907   object, the function zeta, whose structure fixes where its zeros lie." -/

```

```

908
909 /-- A foundation resource is any property of frames the foundation supplies. -/
910 def Resource := Frame → Prop
911
912 /-- (a) Misaddressed: any resource that also holds on a frame where the line property fails
913 cannot, by itself, yield the line property on every frame it covers. -/
914 -- Theorem 63.
915 theorem misaddressed_to_foundation (R : Resource) (hR : R twoPoint) :
916   ¬ ∀ X, R X → LineProperty X :=
917   fun h => refuted_by_witness twoPoint true trivial (fun e => Bool.noConfusion e) (h twoPoint hR)
918
919 /-- (b) Presupposition: a derivation of the value from a reading equivalent to it uses the
920 value's own content; the reading and the value stand or fall together. -/
921 -- Theorem 64.
922 theorem derivation_presupposes {V P : Prop} (leg : P ↔ V) : (P → V) ∧ (V → P) ∧ (¬ V → ¬ P) :=
923   (leg.mp, leg.mpr, fun nv p => nv (leg.mp p))
924
925 /-- (c) Well-addressed: the object decides. A frame settles its own line property, in one
926 direction by a located off-line witness, in the other by its own structure. -/
927 def onLineFrame : Frame := (Unit, fun u => u, fun _ => True)
928 -- Theorem 65.
929 theorem object_decides :
930   ¬ LineProperty twoPoint ∧ LineProperty onLineFrame :=
931   (refuted_by_witness twoPoint true trivial (fun e => Bool.noConfusion e), fun _ _ => rfl)
932
933 /-- THE DEMAND, IN TOTO. Made to the foundation, it is misaddressed and any answer from
934 there presupposes the value; made to the object, it is well-posed and the object answers. -/
935 -- Theorem 66.
936 theorem demand_correctly_addressed :
937   (∀ R : Resource, R twoPoint → ¬ ∀ X, R X → LineProperty X) ∧
938   (∀ {V P : Prop}, (P ↔ V) → (¬ V → ¬ P)) ∧
939   (¬ LineProperty twoPoint ∧ LineProperty onLineFrame) :=
940   (misaddressed_to_foundation, fun leg => (derivation_presupposes leg).2.2, object_decides)
941
942 end DotRH
943
944 namespace DotRH
945 /-- Theorem 67. Attached to the paper's cone: the readings of Parts X and XI collapse onto
946 the hypothesis, and the foundation marks the value [.]. -/
947 def rhCollapse (H : RALi.Hunt) : Collapse H.L.RH := (RALi.Reg5, RALi.D5 H, RALi.hunt_cone H)
948 theorem rh_marked_dot (H : RALi.Hunt) : emitAtFoundation (rhCollapse H) = .dot := rfl
949 end DotRH
950
951 #print axioms DotRH.self_referential_limit
952 #print axioms DotRH.value_marked_dot
953 #print axioms DotRH.dot_is_not_a_verdict
954 #print axioms DotRH.demand_is_ghost_in_register
955 #print axioms DotRH.denial_reenacts_root
956 #print axioms DotRH.external_proof_adds_nothing
957 #print axioms DotRH.refuted_by_witness
958 #print axioms DotRH.line_not_self_grounding
959 #print axioms DotRH.foundation_hardenened
960 #print axioms DotRH.misaddressed_to_foundation
961 #print axioms DotRH.derivation_presupposes
962 #print axioms DotRH.object_decides
963 #print axioms DotRH.demand_correctly_addressed
964 #print axioms DotRH.rh_marked_dot
965
966 /-! ## PART XIII · The six-register cone: Postulate M joined to the apex. -/
967 namespace SixCone
968
969 /-- The hunt with one more reading. The zeros are enumerated by stage, the seed holds, and
970 the hypothesis is, by its definition, the line property of the enumerated zeros; that
971 identification enters as the hypothesis `hDef`, carried by citation as Li's criterion and
972 Newman's equivalence are. -/
973 structure Hunt6 extends RALi.Hunt where
974   α      : Type
975   Z      : α → Prop
976   onL    : α → Prop
977   stage  : α → Nat
978   seed   : TimeLocus.Lstage Z onL stage 0
979   hDef   : L.RH ↔ ∀ z, Z z → onL z
980
981 inductive Reg6 where
982   | five (r : RALi.Reg5)
983   | postulateM
984
985 def D6 (H : Hunt6) : Reg6 → Prop
986   | .five r      => RALi.D5 H.toHunt r
987   | .postulateM => TimeLocus.PostulateM H.Z H.onL H.stage
988
989 -- Theorem 68.
990 /-- The cone over six readings: the five of Part X and Postulate M, every leg an equivalence

```

```

991   with the hypothesis. -/
992 theorem hunt_cone6 (H : Hunt6) :  $\forall r, D6\ H\ r \leftrightarrow H.L.RH :=$  by
993   intro r
994   cases r with
995   | five r => exact RALi.hunt_cone H.toHunt r
996   | postulateM =>
997     exact (TimeLocus.postulateM_iff_line H.Z H.onL H.stage H.seed).trans H.hDef.symm
998
999 -- Theorem 69.
1000 /-- The collapse foundation over six readings, and its mark on the value. -/
1001 def rhCollapse6 (H : Hunt6) : DotRH.Collapse H.L.RH := {Reg6, D6 H, hunt_cone6 H}
1002 theorem rh_marked_dot6 (H : Hunt6) : DotRH.emitAtFoundation (rhCollapse6 H) = .dot := rfl
1003
1004 end SixCone
1005
1006 #print axioms SixCone.hunt_cone6
1007 #print axioms SixCone.rh_marked_dot6
1008
1009 namespace DotRH
1010 -- Theorem 63, general form.
1011 /-- Any resource that also holds on some frame whose line property fails cannot, by itself,
1012   yield the line property on every frame it covers. -/
1013 theorem misaddressed_general (R : Resource) (Y : Frame) (hY :  $\neg$  LineProperty Y) (hR : R Y) :
1014    $\neg \forall X, R\ X \rightarrow$  LineProperty X :=
1015   fun h => hY (h Y hR)
1016 end DotRH
1017
1018 #print axioms DotRH.misaddressed_general
1019
1020 /-! ## PART XIV · The logical form of the hypothesis: the only existence lives in the denial. -/
1021 namespace LogicalForm
1022
1023 -- Theorem 70.
1024 /-- A universal sentence carries no existential import: over an empty zero set the line
1025   property holds vacuously. The hypothesis posits no object. -/
1026 theorem universal_posits_nothing { $\alpha$  : Type} (onL :  $\alpha \rightarrow$  Prop) :
1027    $\forall z : \alpha, (fun \_ => False)\ z \rightarrow$  onL z :=
1028   fun _ h => h.elim
1029
1030 -- Theorem 71.
1031 /-- The denial posits an object: a located zero off the line refutes the hypothesis. -/
1032 theorem denial_posits_a_witness { $\alpha$  : Type} (Z onL :  $\alpha \rightarrow$  Prop) :
1033   ( $\exists z, Z\ z \wedge \neg$  onL z)  $\rightarrow \neg \forall z, Z\ z \rightarrow$  onL z :=
1034   fun (z, hz, off) h => off (h z hz)
1035
1036 -- Theorem 72.
1037 /-- With a decidable line predicate, the hypothesis fails only by a witness: it holds iff no
1038   off-line zero exists. The denial carries the whole existential load. -/
1039 theorem fails_only_by_witness { $\alpha$  : Type} (Z onL :  $\alpha \rightarrow$  Prop) [V z, Decidable (onL z)] :
1040   ( $\forall z, Z\ z \rightarrow$  onL z)  $\leftrightarrow \neg \exists z, Z\ z \wedge \neg$  onL z := by
1041     constructor
1042     · intro h (z, hz, off); exact off (h z hz)
1043     · intro h z hz
1044       exact Decidable.byContradiction (fun off => h (z, hz, off))
1045
1046 -- Theorem 73.
1047 /-- A witness is a finite check: for a decidable predicate on the naturals, one index at which
1048   the check fails refutes the universal, and the check at that index is a computation. -/
1049 theorem witness_is_a_finite_check (bad : Nat  $\rightarrow$  Bool) (n : Nat) (h : bad n = true) :
1050    $\neg \forall m, bad\ m = false :=$ 
1051   fun hall => by rw [hall n] at h; cases h
1052
1053 end LogicalForm
1054
1055 #print axioms LogicalForm.universal_posits_nothing
1056 #print axioms LogicalForm.denial_posits_a_witness
1057 #print axioms LogicalForm.fails_only_by_witness
1058 #print axioms LogicalForm.witness_is_a_finite_check

```

APPENDIX B: THE KERNEL AUDIT, VERBATIM

lean RA_Li_Bridge.lean, exit 0, no warnings, no errors.

```

1 'RALi.bridge_yields_RH' does not depend on any axioms
2 'RALi.bridge_is_keyed' does not depend on any axioms
3 'RALi.no_uniform_bridge' does not depend on any axioms
4 'RALi.li_is_a_bit_stream' does not depend on any axioms
5 'RALi.xi0_retires_iff_unity' depends on axioms: [propext]
6 'RALi.retired_seal_is_sound' does not depend on any axioms
7 'RALi.retired_grade_capped' does not depend on any axioms
8 'RALi.unity_is_keyed' does not depend on any axioms

```

```

9 'RALi.ghost' does not depend on any axioms
10 'RALi.common_phi_is_uniform' does not depend on any axioms
11 'RALi.blind_constructions_die' does not depend on any axioms
12 'RALi.mertens_nonneg' depends on axioms: [propept, Quot.sound]
13 'RALi.unitary_iff_on_line' depends on axioms: [propept, Quot.sound]
14 'RALi.off_line_forces_growth' depends on axioms: [propept, Quot.sound]
15 'RALi.stability_iff_line' depends on axioms: [propept, Quot.sound]
16 'RALi.one_repeats_one' does not depend on any axioms
17 'RALi.the_step_is_the_claim' does not depend on any axioms
18 'RALi.monism_posit_forces' does not depend on any axioms
19 'RALi.finite_never_forces' does not depend on any axioms
20 'RALi.mixed_is_fold_invariant' does not depend on any axioms
21 'RALi.mixed_recurrence_fails' does not depend on any axioms
22 'RALi.full_gives_actual' does not depend on any axioms
23 'RALi.actual_not_full' does not depend on any axioms
24 'RALi.sufficient_time' does not depend on any axioms
25 'RALi.one_involution_constructed' does not depend on any axioms
26 'RALi.timeless_equals_timed' does not depend on any axioms
27 'RALi.monism_closes' does not depend on any axioms
28 'RALi.monism_witness_is_the_claim' does not depend on any axioms
29 'RALi.constructed_monism' does not depend on any axioms
30 'RALi.monism_absent_on_breaks' does not depend on any axioms
31 'RALi.monism_yields_RH' depends on axioms: [propept, Quot.sound]
32 'RALi.monism_on_li_is_RH' depends on axioms: [propept, Quot.sound]
33 'RALi.global_monism_inconsistent' does not depend on any axioms
34 'RALi.anchored_closes' does not depend on any axioms
35 'RALi.anchored_derives_monism' does not depend on any axioms
36 'RALi.lyapunov_is_monism' depends on axioms: [propept]
37 'RALi.closure_grade_is_transport_grade' does not depend on any axioms
38 'RALi.rh_anchor_is_the_claim' depends on axioms: [propept, Quot.sound]
39 'RALi.flow_monotone' depends on axioms: [propept, Quot.sound]
40 'RALi.reality_transport' depends on axioms: [propept, Quot.sound]
41 'RALi.rh_iff_lambda_zero' depends on axioms: [propept, Quot.sound]
42 'RALi.future_closed' depends on axioms: [propept, Quot.sound]
43 'RALi.flowed_record_forgets' depends on axioms: [propept, Quot.sound]
44 'RALi.upstream_is_not_forced' depends on axioms: [propept, Quot.sound]
45 'RALi.hunt_cone' depends on axioms: [propept, Quot.sound]
46 'RALi.one_gap' depends on axioms: [propept, Quot.sound]
47 'RALi.regress_halts' depends on axioms: [propept, Quot.sound]
48 'RALi.nothing_weaker' does not depend on any axioms
49 'RALi.sufficient_closes_all' depends on axioms: [propept, Quot.sound]
50 'Timelocus.bound_iff_line' does not depend on any axioms
51 'Timelocus.time_does_not_bind' does not depend on any axioms
52 'Timelocus.monism_holds_in_counter_world' does not depend on any axioms
53 'Timelocus.timeless_with_seed_binds' does not depend on any axioms
54 'Timelocus.witness_iff_bound' does not depend on any axioms
55 'Timelocus.constructed_at_model' does not depend on any axioms
56 'Timelocus.not_constructible_off_line' does not depend on any axioms
57 'Timelocus.postulateM_iff_line' does not depend on any axioms
58 'Timelocus.postulateM_decides' does not depend on any axioms
59 'DotRH.self_referential_limit' does not depend on any axioms
60 'DotRH.value_marked_dot' does not depend on any axioms
61 'DotRH.dot_is_not_a_verdict' does not depend on any axioms
62 'DotRH.demand_is_ghost_in_register' does not depend on any axioms
63 'DotRH.denial_reenacts_root' does not depend on any axioms
64 'DotRH.external_proof_adds_nothing' does not depend on any axioms
65 'DotRH.refuted_by_witness' does not depend on any axioms
66 'DotRH.line_not_self_grounding' does not depend on any axioms
67 'DotRH.foundation_harden' does not depend on any axioms
68 'DotRH.misaddressed_to_foundation' does not depend on any axioms
69 'DotRH.derivation_presupposes' does not depend on any axioms
70 'DotRH.object_decides' does not depend on any axioms
71 'DotRH.demand_correctly_addressed' does not depend on any axioms
72 'DotRH.rh_marked_dot' depends on axioms: [propept, Quot.sound]
73 'SixCone.hunt_cone6' depends on axioms: [propept, Quot.sound]
74 'SixCone.rh_marked_dot6' depends on axioms: [propept, Quot.sound]
75 'DotRH.misaddressed_general' does not depend on any axioms
76 'LogicalForm.universal_posits_nothing' does not depend on any axioms
77 'LogicalForm.denial_posits_a_witness' does not depend on any axioms
78 'LogicalForm.fails_only_by_witness' does not depend on any axioms
79 'LogicalForm.witness_is_a_finite_check' does not depend on any axioms

```
